

Optimization of the Excitation Pattern for Pulsed-Interleaved MINFLUX Nanoscopy

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Optimization of the Excitation Pattern for Pulsed-Interleaved MINFLUX Nanoscopy

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Eingereicht von: Lars Richter

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Abstract

Super-resolution microscopy has revolutionized the imaging of biological systems. MINFLUX nanoscopy is an extremely photon efficient technique for locating single fluorescent molecules. The method routinely reaches the ultimate limit of 1-2 nm precision localizing organic fluorophores in nanoscopic imaging and nanoscale tracking. MINFLUX determines the location of a single fluorescent molecule from the recorded intensity upon excitation with a light pattern of spatially and temporally separated zeroes of intensity, which needs to be precisely known.

Based on simulations and experimental measurements, this work studies the processes for imaging and alignment of the excitation beam pattern and their influence on the performance of MINFLUX. It is found that the experimental parameters for the image acquisition significantly influence the attainable localization precision. A range of optimum imaging parameters is obtained.

The analyses reveal that MINFLUX is rather tolerant to misalignments of the excitation beam pattern; misalignments of 15 nm produce localization uncertainties smaller than 0.5 nm. An efficient alignment protocol for the excitation beam pattern is presented. Finally, MINFLUX nanoscopy with pulsed interleaved excitation is demonstrated, localizing single nanoparticles with 1 nm precision, using 100-fold fewer photons compared to camera-based techniques.

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List of Symbols

λ	Wavelength of light
τ	Fluorescence lifetime
FOV	Field of view
K	Number of doughnut-shaped excitation PSFs
k_r	Rate constant for radiative transitions
k_{isc}	Rate constant for intersystem crossing
k_{nr}	Rate constant for non-radiative transitions
L	Diameter of MINFLUX excitation beam pattern
N	Total number of photons (signal + noise)
px	Pixel size
S	Singlet state
SBR	Signal-to-background ratio(s)
SNR	Signal-to-noise ratio(s)
T	Triplet state
APD	Avalanche photodiode
Au NP	Gold nanoparticle
BFP	Back focal plane
CPU	Central processing unit
CRB	Cramér-Rao lower bound
CW	Continuous wave
DAQ board	Data acquisition board
EBP	Excitation beam pattern
FWHM	Full width half maximum
GPU	Graphics processing unit, here referring to the graphics card
GUI	Graphical user interface

IR	Infrared
MINFLUX	Minimal photon fluxes
MLE	Maximum likelihood estimator
NA	Numerical aperture
p-MINFLUX	Pulsed-interleaved MINFLUX nanoscopy
PBS	Polarizing beam splitter
PSF	Point spread function
ROI	Area of interest
SIM	Structured illumination microscopy
SLM	Spatial light modulator
SMLM	Single-molecule localization microscopy
STED	Stimulated emission depletion
TCSPC	Time-correlated single photon counting
TEM	Transverse electromagnetic
VPP	Vortex phase plate

1. Introduction

A great majority of us will carry the memory of the day at school when looking through a microscope and sketching some biological structure with pen and paper. This day, few might have asked how small the smallest visible detail is. A question which likely remained unanswered until a physics teacher taught us that there is a fundamental limit found by a scientist called Ernst Abbe.

Abbe famously determined that an optical system can only resolve details down to about half the wavelength of the light used for illuminating the object. Using the most powerful objectives available, this so-called diffraction limit lies in the ranges of 200-300 nm when imaging in the focal plane and about 500-600 nm along the optical axis [1].

In 2014, the Nobel Prize in Chemistry was awarded 'for the development of super-resolved fluorescence microscopy'. The recognized methods, also summarized under the name far-field fluorescence nanoscopy, are revolutionizing biological imaging combining sub-diffraction spatial resolution and low-invasiveness. The key for reaching spatial resolution beyond the diffraction limit is not the optics but the fluorophores. Targeted turning on and off of fluorophores permits that only emitters within a sub-diffraction region are located at the same time [2].

The way the fluorescence is switched on and off identifies two groups of nanoscopy techniques. Coordinate-stochastic methods, also known as single-molecule localization microscopy, use a uniform widefield excitation and camera-based image detection. The fluorophores are switched ON stochastically in time and space allowing to image each emitter individually. Centroid fitting of the single-molecule signals yields the reconstructed, super-resolved image [3, 4]. This contrasts with coordinate-targeted techniques which make use of light-induced OFF-switching of the fluorophores in specific regions of the sample. An illumination pattern with a region of zero intensity depletes the previously excited fluorescent state. The sample is imaged by scanning the excitation and depletion beams [5, 6].

Although conceptually unlimited, the resolution of coordinate-targeted and coordinate-stochastic methods is restricted by the photostability and the photon emission rate of the fluorophores. In practice, both families of nanoscopy techniques feature 20-50 nm lateral and 60-120 nm axial resolution imaging biological structures [7]. Continuous efforts aim to increase the photon budget provided by the fluorophores [8]. Yet, sub-10 nm spatial resolution remains inaccessible for those first-generation nanoscopy methods.

Second-generation nanoscopy employs two concepts to reach the ultimate resolution limit imposed by the size of the fluorophores. Higher spatial resolution can be achieved increasing the number of photons available at a specified sample location. Alternatively, more information is extracted from a given photon budget using prior knowledge of a spatially modulated excitation pattern.

In DNA-PAINT, transient hybridization of short, fluorescently labelled single-stranded DNA sequences enables the blinking at specifically targeted sample locations. Spatial resolution below 5 nm was achieved by significantly increasing the photon budget [9, 10, 11, 12, 13]. MINFLUX employs a modulated excitation pattern with a defined intensity zero. The pattern

is shifted with respect to the sample and the knowledge of the excitation pattern allows localizing fluorophores with 1 nm precision. The method requires 22 times fewer photons for reaching 5 nm localization precision in comparison to camera-based methods [14].

MINFLUX currently provides the best spatio-temporal resolution amongst all nanoscopy techniques [15]. Isotropic resolution in three dimensions was demonstrated [16]. Recently, the method was employed for studying proteins in cells at unprecedented resolution [17]. Despite the power of MINFLUX, its high technical complexity limits the widespread application in microscopy laboratories.

The idea to excite a fluorophore by a modulated intensity pattern around an intensity zero was combined with less complex widefield illumination approaches. Those modulation-enhanced nanoscopy modalities yield a two-fold resolution enhancement compared to single-molecule localization microscopy. [18, 19, 20, 21, 22]. However, they lack the possibility to zoom into the sample.

This work was carried out in the context of a simpler experimental realization of MINFLUX employing pulsed-interleaved excitation [23]. Based on simulations and measurements, optimization strategies for the alignment and the imaging of the excitation beam pattern are studied.

First, it is shown that the parameters used for imaging the excitation beam pattern have a significant influence on the attainable localization precision. The optimum range of experimental parameters is reported. Second, the study reveals that the MINFLUX localization scheme is robust to misalignments of the excitation beam arrangement.

Finally, the performance of pulsed-interleaved MINFLUX is demonstrated in a proof of concept experiment. A graphical user interface was developed to facilitate the data analysis process. Single gold nanoparticles were localized with the theoretically predicted precision of 1-2 nm using a 100 times reduced photon budget compared to camera-based detectors.

2. Fluorescence Nanoscopy and Localization Estimators

This chapter introduces the relevant theory to understand how MINFLUX works, and why it is a groundbreaking innovation in microscopy. In section 2.1 the fundamentals of fluorescence are explained. Next, the basics of fluorescence microscopy are outlined and the subfield of nanoscopy is presented. Section 2.3 describes the underlying concepts in MINFLUX nanoscopy. The chapter concludes by discussing the mathematical estimators applied to benchmark the localization precision.

2.1. What is Fluorescence?

Fluorescence denotes a photophysical process where an atom or molecule emits light due to a transition between two energy levels given that the angular momentum of the system is conserved. Accordingly, the emission of photons as spin-1-bosons requires a change in the total-spin of the electronic-system equal to one [25]. Processes involving photon emission and a violation of angular momentum conservation are called phosphorescence. Both phenomena are referred to as luminescence.

Lifetimes of luminescence transitions differ between nanoseconds for fluorescence and up to seconds for phosphorescence. Here, lifetime refers to the average time τ a molecule remains in the excited state before relaxation. Depopulation of an excited state triggers either a radiative or non-radiative transition with rate constants k_r and k_{nr} , respectively. The rate constants relate to the lifetime τ via

$$\tau = \frac{1}{k} = \frac{1}{k_r + k_{nr}} , \quad (2.1.1)$$

with k_{nr} , summarizing relaxation pathways such as internal conversion.

Whereas in atoms only electronic energy levels are considered, molecules additionally possess vibrational and rotational energy states. The latter will subsequently be neglected as their influence is small in comparison to transitions between electronic or vibrational levels. Most molecules show luminescence in the visible spectrum due to radiative transitions between the electronic Singlet S or Triplet states T . Singlet states exhibit an anti-parallel electron-spin configuration and Triplet states a parallel configuration.

The Jablonski diagram in figure 2.1 illustrates the various photophysical processes including internal conversion and intersystem crossing [26]. In equation 2.1.1, the rate constant for intersystem crossing k_{isc} is neglected as $k_{isc} \ll k_{nr}, k_r$. Hund's rule states that anti-parallel aligned spins are energetically favourable. Therefore, T_1 is sketched below S_1 [27].

Because of internal conversion, that is, for example, a change in vibrational energy, there is an energy difference between photon absorption and emission. Therefore, absorption and emission spectra are shifted. The case of an energy depletion by internal conversion is called Stokes shift while a gain in energy defines the Anti-Stokes shift. All fluorescent

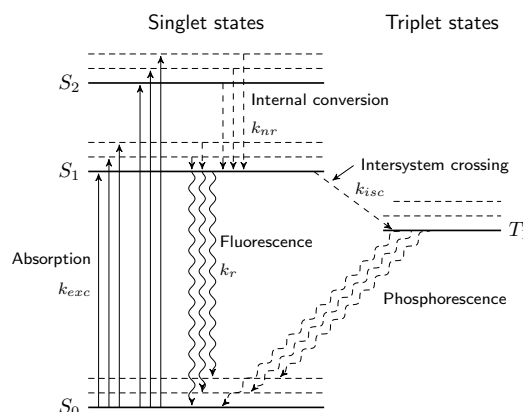


Figure 2.1.: Jablonski diagram. The Singlet-states S_0 , S_1 , S_2 are sketched on the left and Triplet-state T_1 on the right. The photophysical processes are visualized by different arrows. The corresponding rate constants k_{exc} for fluorescence excitation, k_r for radiative decay pathways, k_{nr} for non-radiative decay pathways and k_{isc} for intersystem crossing are indicated.

molecules show a Stokes shift and thus emit electromagnetic waves with less energy, *i.e.* higher wavelengths than the initially absorbed photons.

Figure 2.2 depicts the absorption and emission spectra of the fluorophore Cy3B. Both spectra exhibit symmetric features which are ultimately understood recognizing the Born-Oppenheimer approximation and the Franck-Condon principle [28, 29, 30]

Molecules exhibit fluorescence because of their chemical structure. In the visible spectral range fluorescence emission mainly occurs for compounds with π -electron systems. In contrast, *e.g.* σ -electron systems do not possess a transition dipole moment due to the spherically symmetric charge distribution [32].

The physical and chemical nature of fluorescent dyes causes bleaching and blinking [33]. Bleaching in the context of fluorescent molecules refers to the chemical process of photobleaching denoting the irreversible loss of a dye's ability to fluoresce. This is caused by the destruction of the π -electron system. Photobleaching depends on other molecules in the vicinity of the fluorophores, the irradiance of incident light and the structure of the dyes [34, 35]. Especially reactions with oxygen lead to bleaching [8].

Blinking terms the intermittent emission of single fluorescent systems. Studies of the effect propose different reasons, *e.g.* Triplet blinking, naming intersystem crossing from an excited Singlet state to the first Triplet state [36].

Historically, fluorescence blinking was first observed in atoms due to transitions to metastable states with long lifetimes [37, 38, 39]. Then, Triplet blinking of organic molecules was discovered. Nowadays, as single-molecule spectroscopy became widespread, numerous blinking mechanisms have been identified for molecules, quantum dots, and metallic nanoparticles [40, 41].

2.2. From Fluorescence Microscopy to Fluorescence Nanoscopy

The search for understanding every *detail* of life entails the observation of smaller and smaller structures. For centuries, microscopy has proven to be one of the most useful tools

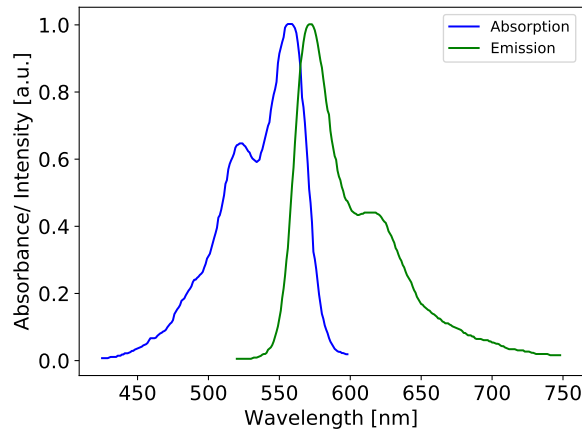


Figure 2.2.: Absorption and emission spectrum of the fluorescent dye Cy3B [31]. The difference in wavelength between absorption and emission peak is caused by the Stokes shift.

in this sense. Especially thanks to the light microscope many groundbreaking advances in life sciences were possible [42, 43, 2].

Employing the phenomena of fluorescence under the objective lens of a microscope offers new possibilities for studying nanoscale structures. In contrast to sole light microscopy, fluorescent dyes add the option to specifically label certain parts of the sample. By that, e.g. distinct biological structures can be non-invasively probed. The Stokes shift allows the wavelength-specific selection of the fluorescence light, reducing background noise and enhancing imaging contrast [44].

From the optics' point of view, two classes of fluorescence microscopes are distinguished. The working principle of both is sketched in figure 2.3. The classical design of a microscope uses the extended excitation of the sample in a wide field of view (*FOV*). Thus, it is termed *widefield microscope*. In widefield microscopy, the excitation beam is focussed on the back focal plane of the objective lens. Consequently, the specimen is illuminated within an area of several hundreds of micrometre at once.

Due to the excitation of a large *FOV*, an image of the sample is acquired at once without the need for scanning. Because of the simultaneous illumination of a large sample volume, background noise is a general problem in widefield microscopy. Moreover, the method is limited when volumetric information from the sample is of interest [45].

The *confocal microscope* has an excitation beam path where the beam is collimated when entering the objective's back focal plane. This leads to the excitation of a nano-to micrometre-sized volume in the sample. Ideally, only within this region fluorescence is emitted. As shown in figure 2.3, the fluorescent light collected by the objective passes a dichroic mirror and is then focussed by a tube lens onto a precisely positioned small aperture, called *pinhole*. The pinhole takes care that no out-of-focus light reaches the detector [46].

The excitation of small, localized sample volumes provides the opportunity to move the focus as desired and thereby section the sample in all three dimensions. Although scanning is slower than capturing a single image of a large *FOV*, the sectioning capability is the main advantage of confocal microscopy compared to the widefield approach. In theory, the

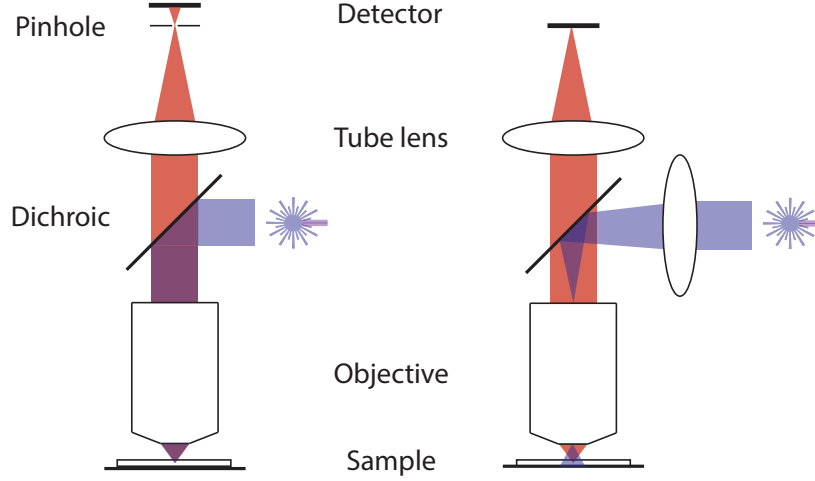


Figure 2.3.: Confocal vs. widefield microscope. The excitation beam is sketched in blue. The fluorescence emission is shown in red. Left: The confocal microscope with a collimated beam entering the objective. Only the fluorescence light passing the pinhole is detected. Right: The widefield microscope with an excitation beam focussed on the back focal plane of the objective.

confocal system allows gaining spatial resolution due to the (de-)convolution of the emission PSF with the aperture function of the pinhole [45]. In practice, this is difficult to achieve because the pinhole is required to be of the size of the emission point-spread function (PSF) which drastically reduces the detected light and hence the image contrast.

In the present section terms like 'large/ small *FOV*' or 'spatial resolution' were mentioned in connection with different fluorescence microscopy techniques. These two quantities describing the performance of a microscope are part of a set of three parameters evaluating the capabilities of any light microscope. Namely, *imaging speed* refers to how fast a certain sample volume is imaged, *spatial resolution* addresses how well the structural details are represented in the image, and *illumination dose* evaluates to how much radiation the sample was exposed to gain the structural information [47, 48].

Ernst Abbe famously derived a physical limit to the achievable spatial resolution of any optical system. This so-called diffraction limit states that it is not possible to discern two objects separated by less than the resolvable distance d . Depending on the exact definition of resolution, the diffraction limit for two laterally distant features is

$$d \approx \frac{\lambda}{2 \text{ NA}} , \quad (2.2.1)$$

λ being the wavelength of light emitted or scattered by the object, and NA the numerical aperture of the light collecting objective lens [1].

At the turn of the century, Eric Betzig, Stefan Hell and Mats Gustafsson proposed and demonstrated that the diffraction limit in far-field optical microscopy can be overcome [4, 49, 50]. Their innovations heralded the era of *super-resolution microscopy*. With reference to the nanometric resolution power, the respective far- and near-field techniques are furthermore summarized under the name *nanoscopy*. Following, the most important classes of nanoscopy techniques are revised.

In **STED** (stimulated emission depletion) microscopy, a gain in resolution is achieved by depopulating the fluorescent state of dyes using a doughnut-shaped de-excitation beam. Assuming the excitation of a large volume followed by the illumination with the doughnut-shaped depletion beam, only in the region of the doughnut centre dyes remain excited and thus emitting light [5]. STED physically reduces the size of the emission PSF and delivers lateral resolution down to 20 nm [6, 51, 52]. As in confocal microscopy, the classical STED microscope requires to scan the sample.

The concept to optically induce a reversible on- and off-switching of fluorescent dyes was adapted in the more general RESOLFT microscopy [53]. Parallelized, multi-colour and 3D extensions of STED and RESOLFT are available [54, 55, 56].

Single-molecule localization microscopy (SMLM) uses the temporal separation of emission events and consecutive centroid fitting for precisely localizing emitters. A super-resolution image of the sample is reconstructed from thousands of sequential raw images which are superimposed and then processed, fitting the position of the emitting species. The challenge in SMLM is to ensure the reliable on- and off-switching of the fluorophores combined with the specific, neither too dense nor too sparse labelling of the sample [51].

SMLM enables imaging with high localization precision which is theoretically distinct from reaching high spatial resolution (*cf.* section 2.4). Nevertheless, SMLM methods as PALM or (d)STORM effectively resolve structures with 20 nm lateral resolution [4, 3, 57]. Recently, DNA-PAINT pushes the limits even further allowing 5 nm spatial resolution in micrometre-sized *FOVs* [9, 13].

Another class of methods providing super-resolution is termed **structured illumination microscopy** (SIM). Here, the sample is illuminated by an interference pattern. The direction of the pattern is rotated with respect to the sample and in the end, about ten raw images, each showing a slightly different illumination pattern, are saved. The striped pattern serves to infer image information from structural details of high spatial frequency. Compared to SMLM, SIM is fast and less invasive as only a small number of images needs to be taken. Yet, its spatial resolution is limited to approximately 100 nm [58, 50, 59].

First reported in 2017, **modulation-enhanced localization microscopy** represents the most recent collection of super-resolution techniques [14]. The idea is to illuminate with a modulated pattern, *i.e.* a pattern leading to strong excitation at certain positions and ideally no excitation at others. The pattern is shifted and knowing the number of photons collected after each shift, the emitter position is *triangulated*. In contrast to all other introduced concepts, the information when no photons were collected is used following the logic that in this case, the emitter must have been at a position of zero excitation intensity.

Relying on minimal photon fluxes, the first method achieving 1 nm localization precision by employing the above scheme was named MINFLUX. In the subsequent section, MINFLUX nanoscopy is introduced in full detail. At this point, it is only of interest to note that three equivalent widefield variants of MINFLUX have been demonstrated. SIMPLE, ROSE and SIMFLUX offer spatial localization precision down to 5 nm, imaging a $(25 \times 25) \mu\text{m}^2$ *FOV* within hundreds of microseconds [18, 19, 20]. ModLoc and SIMPLER translate the aforementioned methods for lateral imaging to the third dimension demonstrating sub-10 nm axial localization precision [21, 22]. A drawback to these widefield approaches is the missing ability to *zoom* into the sample and the excitation by linear polarized light.

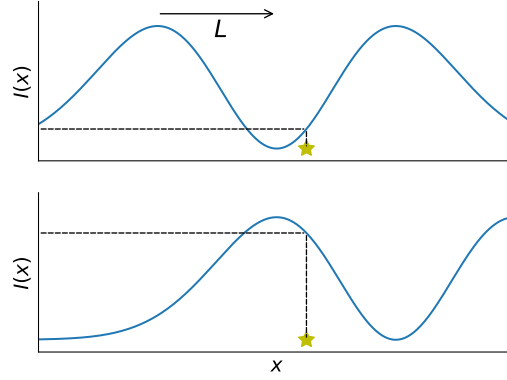


Figure 2.4.: MINFLUX localization concept in one dimension. An emitter (yellow star) is illuminated by a known intensity pattern $I(x)$. The pattern is shifted by a distance L . From $I(x)$ and the detected fluorescence signal before (upper plot) and after (lower plot) the shift of the pattern, the position of the emitter is estimated.

2.3. MINFLUX

MINFLUX establishes the following method for localizing an emitter: A doughnut-shaped excitation beam is shifted with respect to the sample, the photon numbers at each excitation position are counted and a precise image of the excitation beam is taken. Knowing the position of each excitation beam, its shape and the respective photon numbers, the position of the emitter is estimated assuming that the fluorescence signal is proportional to the excitation intensity.

This concept is sketched in figure 2.4 for the one-dimensional case. As depicted, few fluorescence photons will be detected before the shift of the illumination pattern. After the shift, the emitter is more intensively illuminated and will emit stronger. The two collected photon numbers and the known shape of the illumination pattern, provide sufficient information for localizing the emitter.

The power of MINFLUX resides in the modulated excitation. If the emitter is initially located close to the point of zero excitation intensity, the smaller the shift is, the less fluorescence is excited but the more precisely the position is estimated [14]. As it will be derived in subsection 2.4.2, the localization precision of MINFLUX scales as

$$\sigma \sim \frac{L}{\sqrt{N}} , \quad (2.3.1)$$

with N being the total number of photons detected. L describes the shift of the excitation beam pattern (EBP).

Figure 2.5 visualizes a two-dimensional EBP employing $K = 4$ doughnut-shaped excitation PSFs. The doughnut centres are aligned on the corner of an equilateral triangle and its centre. It is important to stress that in MINFLUX nanoscopy the doughnuts excite fluorescence and do not depopulate an excited state as in STED microscopy.

In the first report on MINFLUX nanoscopy, lateral localization precision of 1 nm was

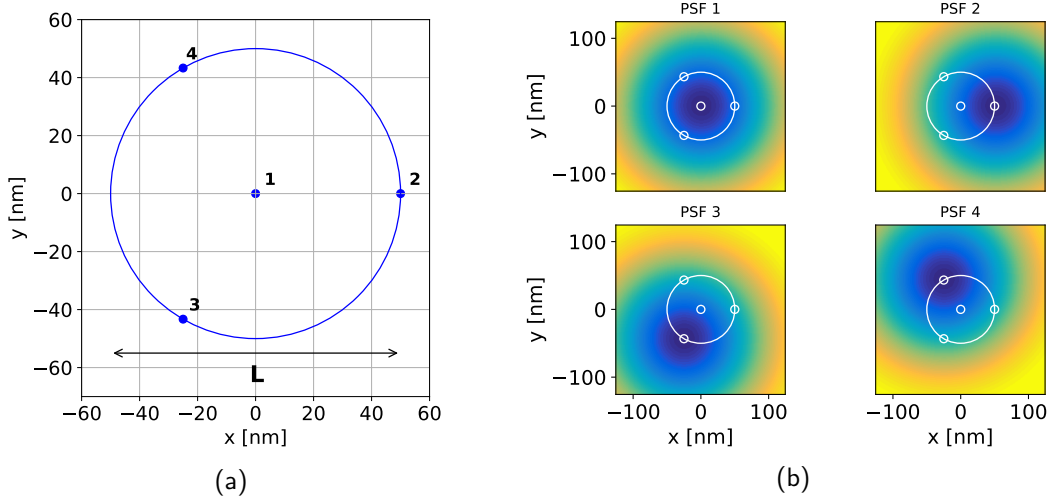


Figure 2.5.: Excitation beam pattern (EBP) for $K = 4$ excitation beams. In (a), the parameter L defining the diameter of the MINFLUX EBP is shown. Subfigure (b) depicts the four doughnut-shaped excitation PSFs. The EBP is visualized in white. For (a) and (b), $L = 100$ nm.

demonstrated [14]. Eilers *et al.* report the so-far best spatio-temporal resolution in nanoscopy localizing fluorescent dyes with 2 nm precision at 400 μ s time resolution [15]. The MINFLUX concept was extended to three dimensions providing isotropic 2 nm localization precision. The axial modulation of the excitation intensity is created by a top-hat beam. Employing a tip-tilt mirror and an electro-optic modulator, FOV s of up to $3 \times 3 \mu\text{m}^2$ are iteratively scanned. The adaptive pattern used for scanning and zooming ensures that the sample is only photoactivated within the respective MINFLUX EBP [16]. Recently, MINFLUX was employed for dual-colour imaging of dense protein structures in cells. 5 nm isotropic localization precision was reported in 3D [17].

Structured light, that is light tailored in space and time to form a desired pattern, serves many applications in manufacturing, optical communication or metrology [60]. With the invention of STED microscopy and a few years later SIM, the use of structured light raised much interest in the microscopy community.

In MINFLUX, a light beam shaped like a torus, or colloquially-speaking a doughnut, is used to create the modulated excitation of the sample. In the focal plane of the objective lens, the doughnut intensity $I(r)$ is described by

$$I(r) = 4 A_0 e \left(\frac{r}{\varrho} \right)^2 \exp \left[-4 \left(\frac{r}{\varrho} \right)^2 \right], \quad (2.3.2)$$

where r is the radial component in polar coordinates, A_0 the peak intensity of the doughnut and ϱ , the peak to peak distance at a fixed polar angle [61, 62]. A simplified form for the central part of the doughnut is found applying a Taylor expansion around $r_0 = 0$

$$I(r) \approx 4A_0e \left(\frac{r}{\varrho} \right)^2 - 16A_0e \left(\frac{r}{\varrho} \right)^4 + \mathcal{O}(r^6). \quad (2.3.3)$$

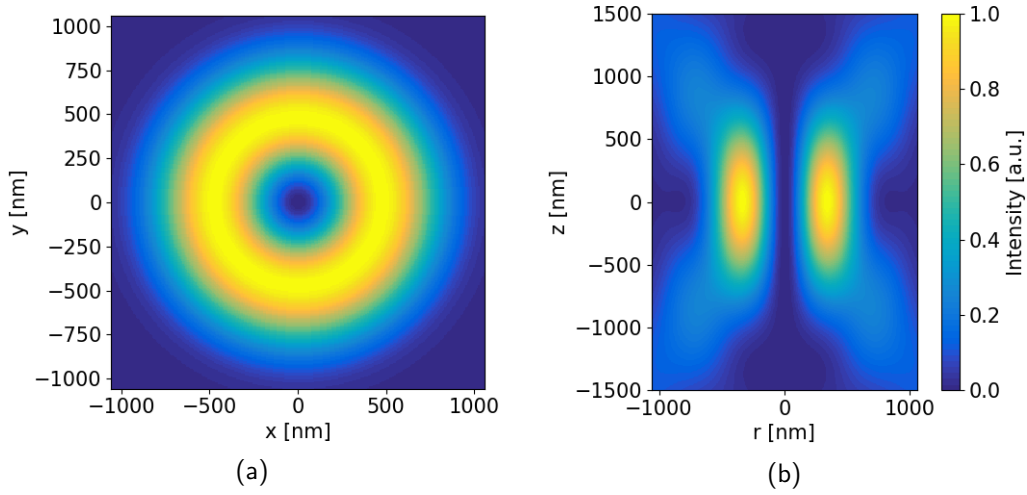


Figure 2.6.: Cross-sections of a doughnut-shaped PSF. (a) xy-slice. (b) xz-slice. Both images share the same colorbar depicted on the right. Adapted from [67].

Terms of $\mathcal{O}(r^6)$ will be neglected as they are ten orders of magnitude smaller than the r^2 contributions.

The prior knowledge of the shape of the excitation beam pattern is crucial to the localization concept of MINFLUX. Several confocal images of the doughnut-shaped excitation PSFs are acquired in a so-called *PSF measurement*. The images are fitted to a grid of 1 nm pixel size. Based on the fitted PSFs, a probability map used for the position estimation is calculated (*cf.* sec. 2.4.1).

Of great experimental interest is the question of how to create a doughnut-shaped focus [63, 64, 65]. Many methods for shaping the amplitude, phase or polarization of light exist. Most conveniently, spatial light modulators or digital micromirrors perform this task [66]. In the first MINFLUX realization, Balzarotti *et al.* used a vortex phase plate (VPP) to create the excitation doughnut [14]. Such a vortex phase plate with topological charge $m = 1$ distorts the phase of the electrical field. The created phase shift increases like a staircase from 0 to 2π leading to constructive interference in the centre of the focussed beam. This point of zero excitation intensity will be referred to as *zero*. Figure 2.6 shows a xy- and xz-cross-section through a doughnut-shaped PSF generated by a VPP.

Similar to a Gaussian emission PSF, a doughnut PSF created by a high-NA objective will only exhibit a radial symmetry if the excitation beam is circularly polarized. In the absence of circular polarization, the doughnut does not possess an area of zero intensity in the beam centre [68, 69].

2.4. Localization Precision and Accuracy

Spatial resolution quantifies how well an optical system discerns two distinct objects. Many different criteria and definitions for spatial resolution have been proposed, most prominently by Lord Rayleigh and Ernst Abbe [1, 70, 71, 72]. For practical purposes, an exact definition is not important as resolution measures are mainly used to compare different instruments or techniques [73].

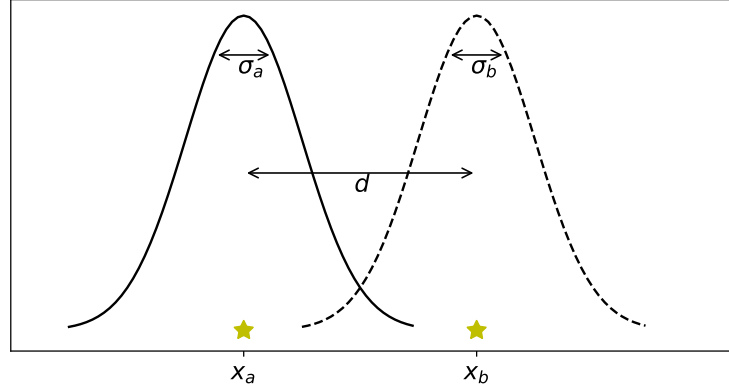


Figure 2.7.: Difference between precision and resolution. The figure depicts two emitters with true positions x_a and x_b separated by a distance d . The location of the emitter is inferred from the measured localization distributions visualized by the two Gaussian functions. The localization precisions σ_a and σ_b are determined by the standard deviation of the respective distribution. In contrast, the spatial resolution evaluates how small d can be in order to still discern the two emitter.

Spatial resolution must be distinguished from *localization precision*. The two terms are often by mistake interchangeably used [74, 75]. Localization precision states how well the position of an object is estimated. Imagine two fluorophores emitting at positions x_a and x_b , spatially separated by a distance d (see figure 2.7). Assuming n measurements of the position of emitter a with results $x_0, x_1, \dots, x_n$, then the standard deviation from the mean $\bar{x}_a$,

$$\sigma_a = \sqrt{\frac{1}{n} \sum_{i=0}^n (x_i - \bar{x})^2} , \quad (2.4.1)$$

defines the localization precision of the respective position estimation.

In contrast, resolution in this example refers to the minimum distance d when it becomes impossible to tell the two emitters apart. If the size of a localization distribution σ_a is associated with the size of an effective emission PSF σ_{PSF} , spatial resolution and localization precision can be related. According to the Rayleigh criterion, $d \sim 2.355 \sigma_a$ [76]. However, high spatial resolution and high localization precision not necessarily coincide, for example, because of inefficient labelling of the underlying structure [77, 78].

Precise localization of an emitter does not ensure that the estimated position is moreover *accurate*. Following the previous example, the estimated position of emitter a , $\bar{x}_a$ can be precise, *i.e.* σ_a is small, but still different from the true emitter position x_a . Thus, the localization would be inaccurate. If the difference

$$\delta_x = \bar{x}_a - x_a , \quad (2.4.2)$$

is zero and thus the accuracy high, the estimator employed for estimating the position is *unbiased*. In contrast to localization precision, the accuracy is not directly determined by separate measurements and thus less sensitive to Poissonian shot noise [76]. Nonetheless,

localization precision is of greater practical use as the true emitter position x_a remains unknown in most experiments.

2.4.1. MINFLUX Position Estimation

In MINFLUX nanoscopy, the location of an emitter is determined using a maximum likelihood estimator (MLE) for the number of photons collected after excitation by each of the K doughnut-shaped PSFs. Knowing how many photons are emitted and holding a good image of the spatio-temporal characteristics of the EBP allows estimating the emitter position. The shape of the EBP and the photon counts are experimentally determined. A joint probability function is constructed, differentiated and set to zero to obtain the maximum likelihood estimate. The latter will tell the most likely emitter position $\hat{\bar{r}}_m$ satisfying the derived statistical model under the condition of the measured photon collection.

In the case of MINFLUX, the conditional probability distribution is given by

$$P(\bar{n}|N, \bar{r}_m) = \frac{N!}{n_0! \cdots n_{K-1}!} \prod_{i=0}^{K-1} p_i(\bar{r}_m) , \quad (2.4.3)$$

n_i being the photons detected after the i^{th} excitation, $N = \sum^K n_i$ its sum and $p_i(\bar{r}_m)$, the probability of detecting these n_i photons with the emitter positioned at $\bar{r}_m = (x_m, y_m)$. The $p_i(\bar{r}_m)$ are determined by

$$p_i(\bar{r}_m) = \frac{\lambda_i(\bar{r}_m)}{\sum_{j=0}^{K-1} \lambda_j(\bar{r}_m)} , \quad (2.4.4)$$

with the $\lambda_i(\bar{r}_m)$ provided by the four fits of the excitation PSFs obtained in the PSF measurement.

The conditional probability distribution as a function of unknown parameter $\bar{r}_m$ defines the likelihood function $\mathcal{L}(\bar{r}|\bar{n}) = P(\bar{n}|N, \bar{r}_m)$ [79]. Thus, the MLE is given by

$$\hat{\bar{r}}_m^{MLE} = \arg \max \mathcal{L}(\bar{r}|\bar{n}) , \quad (2.4.5)$$

which shows that the MLE is the parameter which maximizes the likelihood function. Out of convenience, a modified log-likelihood function l will be used within the scope of this thesis,

$$\ln(\mathcal{L}(\bar{r}|\bar{n})) \sim l(\bar{r}|\bar{n}) = \sum_{i=0}^{K-1} n_i \ln(p_i) . \quad (2.4.6)$$

The emitter position $\bar{r}_m$ which maximizes the (log-)likelihood function is the same as the logarithm is a monotonically increasing function.

The MLE is of great practical use as it is easily computed for the majority of estimation problems. Besides a few exceptions, it is the estimator providing minimum variance, *i.e.* best estimation efficiency [79]. As pointed out by the last sentence, the MLE is not always the best choice. MINFLUX estimations with small photon numbers N , for example, suffer from low localization precision due to the bias intrinsic to the MLE.

2.4.2. Estimator Lower Bounds

The MLE serves to estimate the position of an emitter in a MINFLUX experiment but does not readily report the localization precision. Finding an estimate for the localization precision is possible calculating the Cramér-Rao lower bound (CRB). The CRB defines the lower limit for the variance of an unbiased estimator.

When the MLE defines the most likely position in a MINFLUX estimation, the CRB determines the point(s) of steepest slope for a likelihood function telling where the estimation by the MLE is the most precise. Mathematically, this means calculating the second derivative with respect to the model parameter of interest. For a two-dimensional problem as in MINFLUX, it is convenient to employ the Fisher information matrix $\mathcal{I}$ to calculate the CRB. Balzarotti *et al.* outlined the important steps for this derivation in the case of MINFLUX [61]. For simplicity, only the Fisher information matrix for the MINFLUX estimation in 2D is stated

$$\mathcal{I} = N \sum_{i=0}^K \frac{1}{p_i} \begin{bmatrix} \left(\frac{\partial p_i}{\partial x} \right)^2 & \frac{\partial p_i}{\partial x} \frac{\partial p_i}{\partial y} \\ \frac{\partial p_i}{\partial y} \frac{\partial p_i}{\partial x} & \left(\frac{\partial p_i}{\partial y} \right)^2 \end{bmatrix}, \quad (2.4.7)$$

where the p_i again denote the normalized detection probabilities and $(\partial x, \partial y)$, are the partial derivatives with respect to the x- and y-coordinates of emitter position space.

Knowing the Fisher information matrix, the CRB for the covariance matrix Σ is found

$$\Sigma(\bar{r}_m) \geq \Sigma_{CRB}(\bar{r}_m) = \mathcal{I}^{-1}, \quad (2.4.8)$$

and consequently, the scalar CRB for the localization precision is derived

$$\sigma_{CRB} = \sqrt{\frac{1}{2} \text{tr}(\Sigma_{CRB})}, \quad (2.4.9)$$

where the trace over the inverse Fisher information matrix Σ_{CRB} is taken. σ_{CRB} states the mean of the standard deviations in x and y to ease the comparison of the MINFLUX localization precision to that of camera-based localization techniques. The latter provides isotropic localization precision, *i.e.* the same achievable precision in x and y. In MINFLUX, the localization precision is generally not isotropic [14].

The MLE does not always attain the CRB but it asymptotically converges to it [79]. This property implies for MINFLUX estimations that for small photon numbers or small signal-to-background ratios (SBR), the MLE does not reach the lower limit yielding the highest localization precision. For this reason, alternative estimators with numerically unbiased precision are used in such case [61].

For the MINFLUX EBP with four excitation doughnuts introduced in section 2.3, the highest localization precision for an emitter at the centre of the EBP is

$$\sigma_{CRB}(\bar{r}_m = 0) = \frac{L}{2\sqrt{2N}} \left(1 - \left(\frac{L}{d} \right)^2 \right)^{-1}. \quad (2.4.10)$$

Notable properties are the dependence on the inverse photon number as known for SMLM, and the linear relation to L . The latter implies using smaller EBPs for reaching higher localization precision. Step-wise scanning and zooming into the sample by sequentially reducing L was realized by Gwosch *et al.* and is relevant for the imaging of large $FOVs$

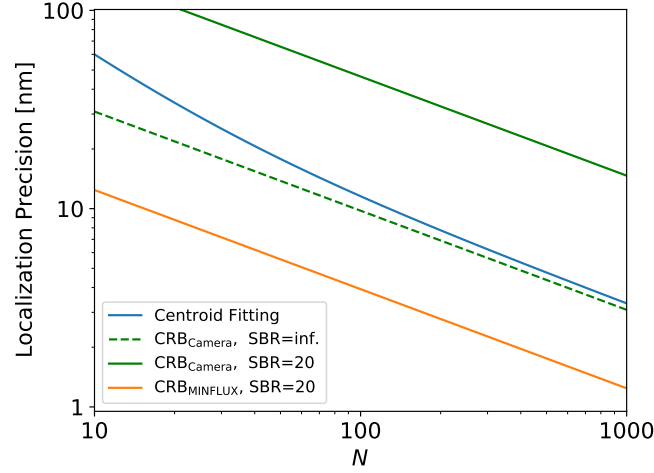


Figure 2.8.: Comparison of estimator lower bounds. Plot of the CRB for a camera detector with a pixel size $px = 100$ nm, $\sigma_{\text{PSF}} = 100$ nm and $K = 9 \times 9$ pixel. The $\text{CRB}_{\text{Camera}}$ is shown for SBR equal to 20 and infinite SBR . The $\text{CRB}_{\text{MINFLUX}}$ was computed assuming an EBP as defined in figure 2.5, an SBR of 20 and the emitter placed in the centre of the EBP. The standard error for centroid fitting is plotted accordingly for the same set of parameters and $b = 1$.

[16].

To compare the localization precision reached by centroid fitting with the CRB obtained in MINFLUX, the CRB for camera-based localization is derived. Using formula 2.4.7 for the Fisher information, the conditional probability locating an emitter with an array-like detector must be known to obtain the CRB. For an instructive derivation, the reader might refer to *p.* 23 *ff.* in [61].

In contrast to MINFLUX, the derived detection probability p_i corresponds to the i^{th} detector's pixel out of the total number of pixel, K . Consequently, the sum to calculate the Fisher information matrix has to be taken over all K pixel.

In 2002, Thompson *et al.* presented a theoretical analysis on centroid fitting deviating the well-known formula for the corresponding standard error on the mean,

$$\langle (\Delta x)^2 \rangle = \frac{\sigma_{\text{PSF}}^2 + px^2/12}{N} + \frac{8\pi b \sigma_{\text{PSF}}^4}{px^2 N^2}, \quad (2.4.11)$$

with σ_{PSF} , being the standard deviation of the emission point-spread function, px the pixel size of the detector array, N the total number of photons collected, and b the standard deviation of the background [80, 81]. Formula 2.4.11 shows that the localization precision in centroid fitting is up to first-order limited by the number of photons from the emitter collected by the detector. Especially in case of a homogeneous background contribution (*i.e.* $b \approx 0$), the $1/\sqrt{N}$ -dependency governs the error in localization Δx .

Apart from the work by Thompson and co-workers, two more approaches theoretically analysing the localization precision shall be mentioned. Bobroff relates the error in the position measurement directly to the error in the fitted parameter for the centroid position evaluating the least-squares estimator [82]. Similarly Kubitschek *et al.* access the problem,

analysing the precision of the least-squares centroid fit [83, 84].

Both of the latter methods inherit the disadvantage that pixelation noise by the detector is disregarded. Additionally, they focus on the fitting of a measured signal instead of following purely physical arguments as Thompson *et al.* do.

Figure 2.8 depicts a plot of the CRB of MINFLUX and for camera-based localization as a function of the total photon number N . The standard error for centroid fitting defined by equation 2.4.11 is shown. Comparing the CRBs visualizes the power of MINFLUX nanoscopy regarding localization precision. The centroid fit precision converges for large N to the camera CRB with infinite SBR ; a proof that Thompson *et al.* constructed a lower-bound estimator. Furthermore, in agreement with the theory, the least-squares estimator for centroid fitting and the MLE-based CRB for the camera coincide in good approximation for large N as the MLE is then well described by a Gaussian model and thus becomes equivalent to a least-squares estimator [85].

3. Pulsed-Interleaved MINFLUX

Pulsed-interleaved MINFLUX (p-MINFLUX) proposes a novel experimental realization of MINFLUX redesigning its original measurement scheme [23]. Central to the concept is the (de)multiplexing of the fluorescence excitation and emission. A beam splitting and recombination arrangement produces an EBP with four temporally interleaved excitation pulses. The measured fluorescence signal is demultiplexed by time-correlated single photon counting (TCSPC).

Figure 3.1 visualizes one *multiplex cycle* of a p-MINFLUX experiment. The four temporally interleaved laser pulses excite fluorescence decaying with a lifetime τ . Each of the pulses corresponds to the excitation at a spatially distinct EBP centre position. The arrival times t_i of the pulses are experimentally determined by the difference in optical path length between beam splitting and beam combination optics. The repetition rate of the pulsed laser dictates the length T of a multiplex cycle. After several thousand to millions of multiplex cycles, the relative photon arrival times corresponding to each of the K excitations are summed up, providing the $\bar{n}$ -array for the MINFLUX estimation. The picosecond excitation pulses combined with time-correlated single-photon detection give furthermore access to fluorescence lifetime information from the localized emitter [86].

The classical MINFLUX scheme employs an electro-optic modulator to shift the doughnut PSF to the four EBP centre positions. One multiplex cycle is reported to last 125 μs . For a pulsed laser with 40 MHz repetition rate, a p-MINFLUX multiplex cycle is 25 ns long. This gain in speed promises to further improve the temporal resolution in MINFLUX nanoscopy.

Within the scope of this thesis, an existing p-MINFLUX setup was used to find a reliable protocol for the alignment of the four doughnut excitation PSFs. The setup and the control software were equipped with new elements enhancing the functionality and usability. The existing workflow for the data analysis was extended and a graphical user interface was introduced.

3.1. Optical Setup

Figure 3.2 illustrates the experimental setup showing the main parts of the p-MINFLUX nanoscope. From excitation light source to fluorescence detection, the light passes the following components. A supercontinuum laser (Fianium SC 400-4, NKT Photonics) emits pulses of ~ 6 ps width (FWHM) at a repetition rate of $1/T = 40$ MHz. The emission spectrum ranges from $\lambda \approx 420$ nm to $\lambda \approx 2000$ nm.

The laser beam is expanded and collimated. Infrared light with $\lambda > 1064$ nm is filtered by a dichroic mirror (BLP01-1064R-25, Semrock Inc.). A fraction is sent to a custom-built, fast photodiode sensing the arrival of the laser pulses. The majority of the far-infrared light is dumped. Near-infrared light is split from the main beam (T750SPXRXT-UF1, Chroma Technology Corporation) and coupled into a multimode fiber. The remaining light is filtered for the red spectral components around $\lambda = 642$ nm (ZET642/20x, Chroma Technology Corporation). The light intensity is controlled using a flipper with a mounted ND filter

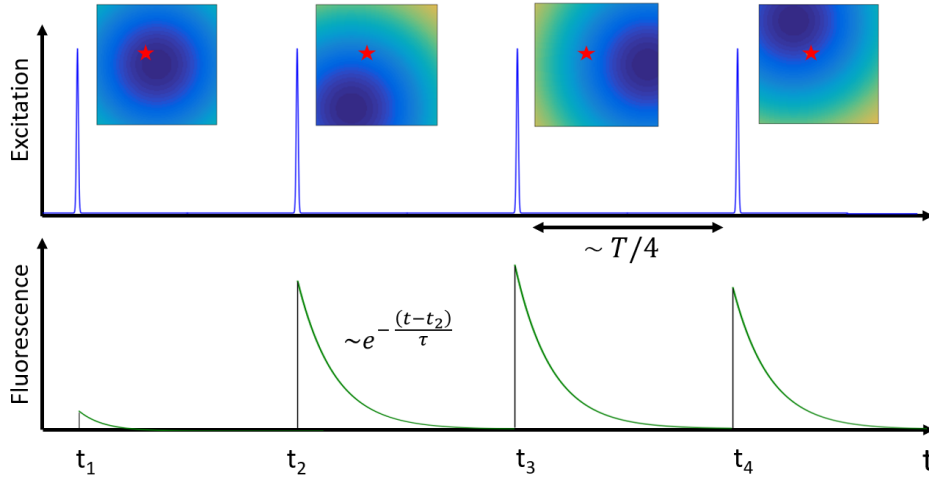


Figure 3.1.: p-MINFLUX excitation scheme. Spatially- and temporally-separated laser pulses excite an emitter (red star). The detected fluorescence scales with the relative position of the emitter to the zero of the excitation pulses. Excitation pulses at t_i , exponential fluorescence decay with lifetime τ and repetition rate $1/T$ of the pulsed laser.

(MFF101/M and NDK01, Thorlabs Inc.).

The red light is split by a polarizing beam splitter (PBS) (CCM1-PBS251/M, Thorlabs Inc.). The two beams have orthogonal polarization states. Each of the beams is sent to a non-polarizing beam splitter (CCM1-BS013/M, Thorlabs Inc.). This beam splitting system generates two pairs of beams with each pair sharing the orthogonal linear polarization. The four beams are coupled into polarization-maintaining fibers (PM630-HP, Thorlabs Inc.). Each of the fibers has a different optical path length $l_i \approx i \times l_1$, with i running from 1 to 4 and $l_1 \approx 1.3$ m, the length of the shortest fiber. The light coupled out of the fibers carries linear polarization. It is guided through dielectric mirrors M_c (POLARIS-K1S4 and BB1-E02-10, Thorlabs Inc.) to the beam combining system. The four beams are combined by three beam splitters. The axes of linear polarization are matched turning the fiber out-couplers (60FC, Schäfter + Kirchhoff GmbH). The four overlapping beams share the same beam path but have distinct temporal characteristics with an offset of $T/4$ in the timing of the pulses.

The recombined beams are focussed by a lens L_1 ($f = 300$ mm, AC254-300-A-ML, Thorlabs Inc.) to a plane conjugated to the focal plane of the microscope objective. Lens L_1 is the first out of four identical lenses forming a double $4f$ -system. Between L_2 and L_3 a VPP (V-633-10, Vortex Photonics) is placed in the conjugated plane to the back focal plane (BFP) of the objective. The VPP creates the doughnut-shaped focus in the focal plane of the oil-immersion objective (CFI Plan Fluor 100 $\times$, NA = 1.4, Nikon Instruments Inc.).

The focussed beam is circularly polarized by the combination of a $\lambda/2$ - and a $\lambda/4$ -waveplate (WPH05M-633 and WPQ05M-633, Thorlabs Inc.) which are positioned in front of L_3 . The four temporally-separated pulses share the doughnut excitation pattern. To align the four doughnut centres to the desired EBP positions, the mirrors M_c are employed. They are placed in the conjugate plane to VPP and BFP and allow to spatially shift the doughnut-shaped focus by changing their tilt angles.

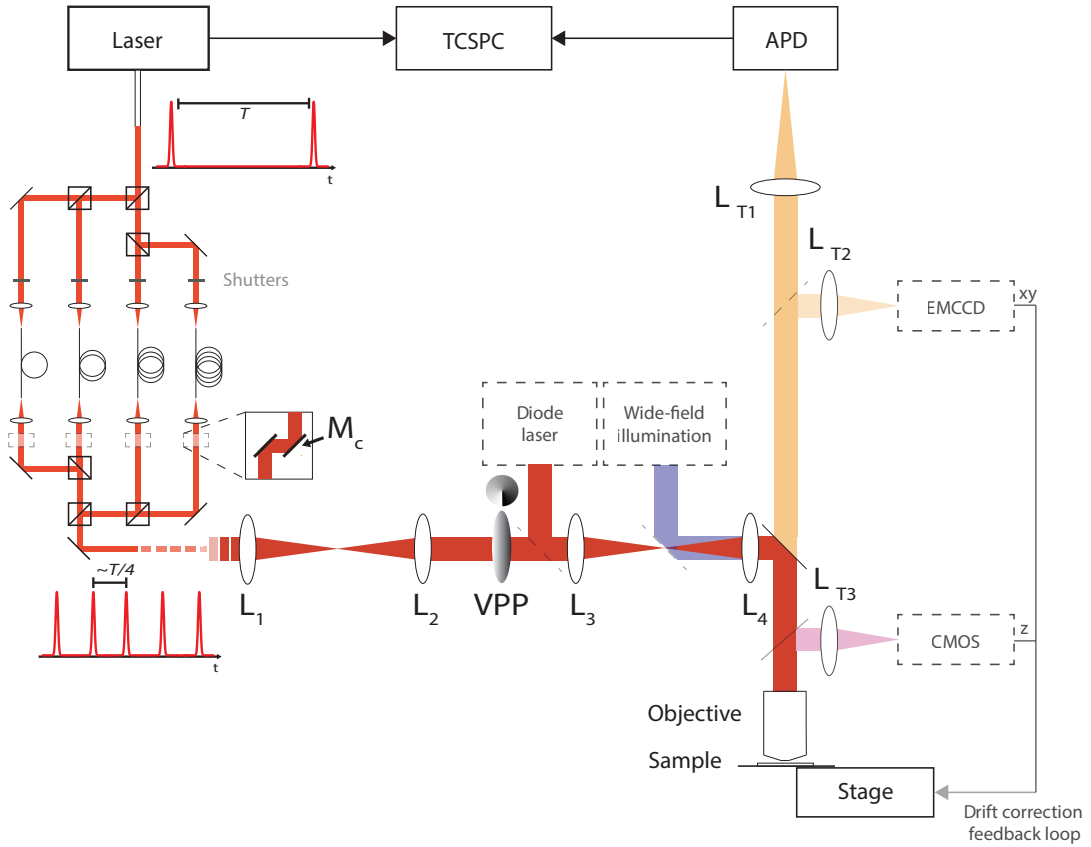


Figure 3.2.: Illustration of experimental p-MINFLUX setup. The pulsed laser light passes a beam splitting system consisting of three beam splitters. Guided by optical fibers, the four created beams experience each a different delay such that they have a temporal offset of $\sim T/4$ after recombination. A double $4f$ -system and a VPP lead to the excitation of the sample by doughnut-shaped PSFs. The fluorescence is detected by an APD which is connected to a TCSPC unit. Sketch adapted from [23].

The light emitted by the sample is collected by the objective. It passes a dichroic mirror (ZT647RDC, Chroma Technology Corporation) which reflects the excitation beams. Four notch filter block the excitation light from the four red excitations beams ($2 \times$ ZET642NF, Chroma Technology Corporation), the blue widefield excitation (ZET473NF, Chroma Technology Corporation) and the reflected infrared beam (NF808-34, AHF analysentechnik AG). The fluorescence emission is focussed by a tube lens (AC254-50-A-ML, Thorlabs Inc.) onto an avalanche photodiode (APD) (SPCM-AQR-13, PerkinElmer Optoelectronics). The APD detects photons with 90 % quantum efficiency at $\lambda = 650$ nm. It provides a single photon timing resolution of 350 ps (FWHM) and typically exhibits 150 dark counts per second.

The digital signal from the fluorescence photons is sent from the APD to a TCSPC unit (PH300 with router PHR800, PicoQuant). The internal timer of the TCSPC unit correlates the photon detection signal from the APD to the signal from the photodiode which senses the arrival of the excitation laser pulse. It returns for every counted photon an absolute time relating to the begin of the measurement and a relative time which was elapsed since the last laser pulse was detected at the photodiode. The temporal resolution of the TCSPC unit is 8 ps and the timing precision < 12 ps. The dead time is reported to be < 95 ns.

The optical system providing z-stabilization employs the near-infrared light coupled into a multimode fibre. After attenuation and clean-up (ET705/72M and T750SPXRXT-UF1, Chroma Technology Corporation), the IR light is guided to the objective. The collimated beam enters the objective at the edge of the back focal aperture. Due to total internal reflection from the sample-coverslip interface, the IR light leaves the objective parallel to the incident IR beam. The reflected light passes a tube lens (AC254-150-A-ML, Thorlabs Inc.) and is focussed on a CMOS camera (ThorCam DCC1240C, Thorlabs Inc.). Through calibration, the z position of the slide can be tracked with a precision of 10 nm and actively corrected [87].

The lateral stabilization of the sample is realized tracking bright green fluorescent beads (FluoSpheres, 0.2 μ m, Nile Red, Thermo Fisher Scientific). As sketched in figure 3.2, a wide-field configuration is employed for illuminating the beads at $\lambda = 473$ nm (GEM 473, Laser Quantum). The fluorescence emission is reflected by a dichroic mirror (ZT532/640RPC-UF, Chroma Technology Corporation) and filtered (ZET473NF, Chroma Technology Corporation). The light is focussed by a tube lens (AC254-300-A-ML, Thorlabs Inc.) and imaged by an EMCCD camera (iXon DV887, Andor). The processing of the data for the xy-stabilization will be explained in the following section.

An auxiliary confocal system consisting of a CW diode laser (Lambda mini Evo, RGB Photonics GmbH) emitting at $\lambda = 640$ nm was added to the setup to provide an excitation with a Gaussian beam profile. Using a pellicle beam splitter (BP108, Thorlabs Inc.), the collimated laser light is coupled into the second 4f-system between VPP and L₃ exciting at the same position as PSF 1.

3.2. Setup Control

The core of the instrument control system is a data acquisition board (DAQ board) (ADwin Gold II, Jäger GmbH) which serves as analogue-digital and digital-analogue converter. The DAQ board allows programming its output channels for the direct control of the connected devices from a computer. The DAQ board receives and processes the raw photon count data from the APD which are used for creating an image from the confocal scan. Additionally, it connects to the piezo controller and the shutters (MediaLas STP 8xl, spectrabeam.de)

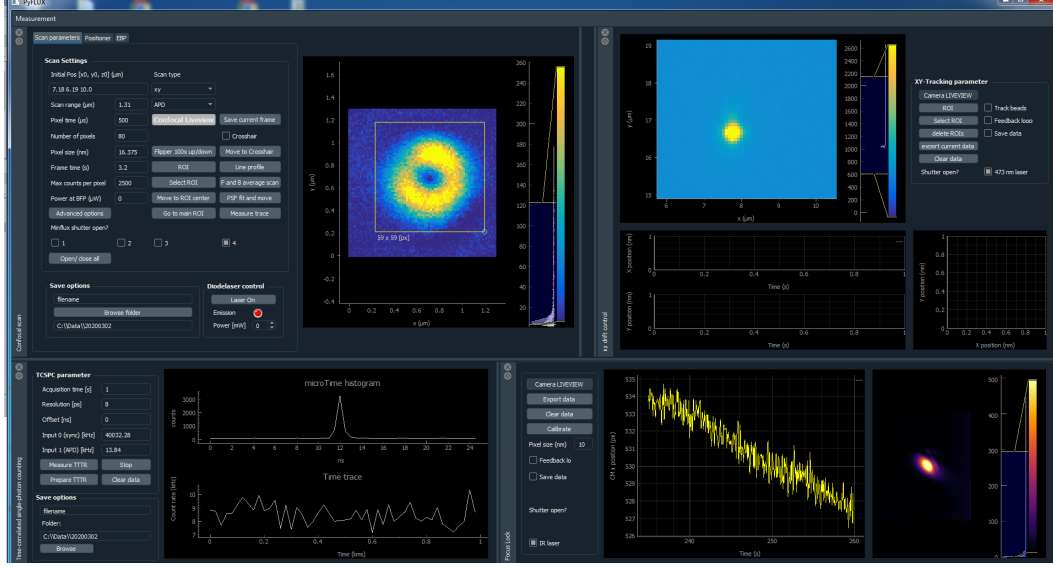


Figure 3.4.: PyFLUX user interface. Modular design with data acquisition modules on the left and xyz-stabilization modules on the right. The confocal scan module on the top-left depicts a live image of a doughnut-shaped PSF. A bright fluorescent bead is shown in the live image for the lateral drift correction system on the top-right. On the bottom-right, the focus lock module depicts an image of the reflected IR beam and a graph visualizing the current z-drift. The TCSPC module on the bottom-left shows in the upper graph a histogram of the relative photon arrival times and in the lower diagram of the absolute photon arrival times. Here, the sample was excited by only one of the four excitation beams.

GUI modules and two measurement modes. Each of those six components runs in a separate thread allowing smooth data acquisition without interferences between the instruments. The modular GUI design of PyFLUX is shown in figure 3.4.

Controlling many devices using one common programming language requires drivers which translate PyFLUX's Python commands to the respective machine code of the instrument. Many manufacturers provide dynamic-link libraries for the customized control of their instruments. Those .dll files allow using pre-defined commands from the library to execute a task on the device. The advantage is that it is a general framework independent of the exact communication protocol of the device. Based on a .dll library, a Python driver can be written employing a C compatible wrapper, for example, ctypes [88].

In the case of the p-MINFLUX setup, .dll libraries for most of the instruments exist. Custom Python drivers were written for the TCSPC unit and the EMCCD camera. The CW diode laser only allows serial communication via USB why a corresponding driver was created using the PySerial package [89]. Table A.1 references the respective libraries and drivers of all the instruments of the p-MINFLUX setup.

Focussing of the fluorescence emission by a tube lens with 150 mm focal length leads to $75\times$ magnification of the image. Combined with the $175\text{ }\mu\text{m}$ diameter of the active area of the APD, this causes that only light from a $\sim 4\text{ }\mu\text{m}^2$ -sized region of the sample is detected, effectively making the APD to function as detector with pinhole. Hence, pulsed laser excitation, piezo stage and APD are regarded to form a *confocal microscope*. The

shape of the excitation PSF in the focal plane of the objective is determined by the optical components along the double $4f$ -system. A badly aligned excitation PSF might show a tilt in xz -coordinates or a focus spot in xy larger than give diffraction-limited. The excitation by deteriorated doughnut-shaped PSFs would prevent attaining high localization precisions across the MINFLUX EBP.

For alignment of the excitation PSF, it is convenient to image the scattering signal from Au NPs by an auxiliary photodiode (PDA36A, Thorlabs Inc.). Scanning and imaging the scattering PSF in xyz , the optical components are adjusted to ensure a diffraction-limited and non-tilted illumination of the sample. During the alignment step, the VPP is removed from the beam path. Employing the APD, the emission PSF of Au NPs is aligned guided by the same criteria as for a well-aligned excitation PSF. The xyz -position of the APD is adjusted carefully to maximize the amount of light incident on the APD. Again the confocal image in xy is compared to the diffraction-limited Gaussian PSF.

The polarization state of the excitation beams is checked to assure circular polarization permitting excitation PSFs with a true zero. We use a custom-built tool consisting of a photodiode and a Glan-Thompson polarizer which is turned at approximately 20 Hz by a motor. The sensed voltage at the photodiode is amplified and observed on an oscilloscope. Depending on the incoming polarization state, the temporal course of the signal shows no modulation for circular or strong modulation for linear polarized light. The modulation is best quantified evaluating the contrast C ,

$$C = \frac{U_{\max} - U_{\min}}{U_{\max} + U_{\min}}, \quad (3.2.1)$$

where $U_{\max/\min}$ denote the maximal/ minimal voltage measured by the oscilloscope. The coupling efficiencies of the four polarization-maintaining fibers were optimized to increase the degree of linear polarization at the out-coupler. The lateral and angular position of the $\lambda/2$ - and $\lambda/4$ -waveplate were adjusted to minimize the contrast and obtain circular polarization. Table A.2 lists the contrast and coupling efficiencies measured after the alignment. Reaching 2-3 % contrast was aimed.

To characterize the temporal response of the pulsed-interleaved excitation scheme, the impulse response function was measured using a silver-coated glass slide. Figure 3.5 depicts the precise timing of the four excitation pulses. The reflection of light occurs on the femtosecond timescale which is three to four times faster than the emission of fluorescence [90]. These distinct interaction times enable the evaluation of the temporal response of the experimental system with sub-nanosecond time resolution. The relative arrival times of the four pulses are found to be $t \approx 1.6, 7.9, 14.3$, and 21.1 ns. This demonstrates that the delay between two successive pulses is approximately $T/4 = 6.25$ ns.

The xyz -stabilization requires prior calibration as the setup control software calculates in px -units of the respective cameras whereas the piezo stage is moved according to an applied voltage. The usual procedure sees a determined movement in x , y or z , set at the piezo controller and the measurement of the shift in the camera image. This is repeated moving the stage by many different distances to fit a proportionality between the set movement of the stage and imaged movement of the fiducial marker or the reflected IR spot.

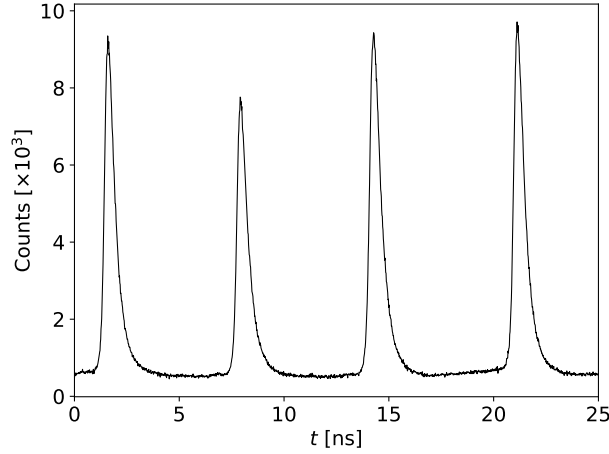


Figure 3.5.: Impulse response function. The reflection of the interleaved laser pulses from a silver-coated glass slide is measured to investigate the timing of the four pulses. The relative arrival times of the four pulses are found to be $t \approx 1.6, 7.9, 14.3,$ and 21.1 ns.

3.3. Measurement Protocol

Two measurements need to be performed to realize a MINFLUX estimation. In both, PSF measurement and MINFLUX measurement, Au NPs and fluorescent beads are used for imaging or tracking. Both species are immobilized on the coverslip of a glass slide with a flow chamber. The immobilization procedure is reported in section A.1.1 of the appendix. The employed 60 nm Au NPs are synthesized according to a protocol developed by Bastús *et al.* [91].

The APD used for the fluorescence photon detection is directly connected to the internal counter of the DAQ board. The counts per set time interval provide information about how many photons were detected at a certain stage position. Moving the stage to another position and counting the photons enables acquiring an image by iterative scanning of a set FOV_{PSF} . Due to the inertia of the piezo responding to the driving signal, scanning along one axis results in a different image than a scan in the opposite direction. The offset between *forward* and *backward* image disappears with lower scanning speeds. If only one image of a sample is of interest this detail is of minor importance and only the forward image is saved. However, when knowing the absolute position of an emitter is relevant, forward and backward image have to be acquired and analysed.

During the **PSF measurement**, confocal images of the four doughnut excitation PSFs are acquired. A field of view of 5-10 μm is scanned using a pixel size of 10-20 nm. A Au NP is selected from the image and centred. The Au NP is imaged in a FOV_{PSF} of 400-600 nm with $px_{\text{PSF}} = 6-10$ nm. The exposure time per pixel was set to 500 μs with a laser power of about 10 μW for the excitation pulses, measured at the back focal aperture of the objective. Before the scan starts the shutters of the first excitation beam, the IR and the widefield beam are opened. The focus lock and xy-stabilization systems are initialized. The two systems actively correct the position of the piezo stage for drift during the PSF measurement. A fixed number of images is acquired for every excitation doughnut PSF before its shutter is closed

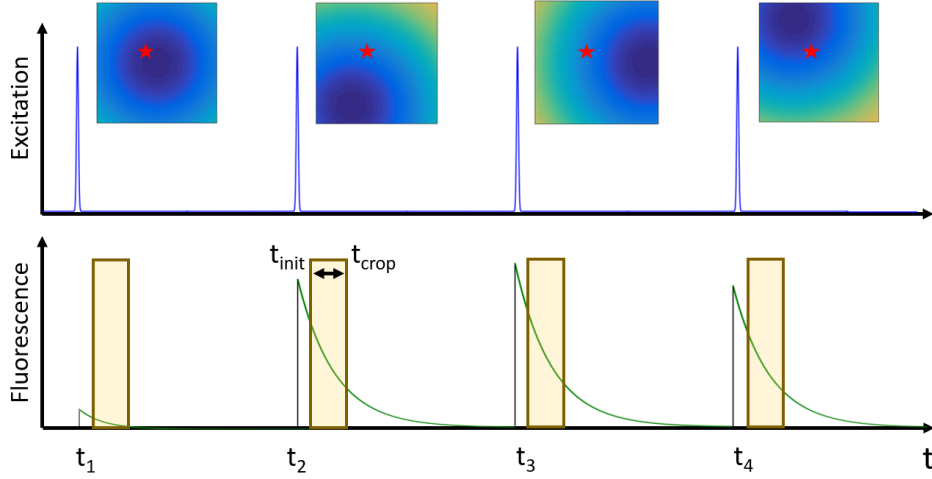


Figure 3.6.: Nanosecond time windows for relative photon arrival times. Only photons counted during the four windows $[t_{\text{init}}, t_{\text{crop}})$ are considered for the estimation of the emitter position.

and the next is opened. This procedure is repeated until all four doughnuts were imaged.

The TCSPC data are collected during the **MINFLUX measurement**. This is carried out while actively correcting for drift in x , y and z . The measurement starts by selecting an area of interest (ROI) from the confocal scan. The algorithm implemented in PyFLUX performs a forward and a backward confocal scan of the set ROI. The emitter position is fitted by a Gaussian in each of the two images. The intermediate position between the two fitted centroids is used as a reference point. The piezo stage is positioned to match the reference point and the zero of excitation PSF 1. To reduce the offset between forward and backward image and allow proper positioning of the sample for the MINFLUX measurement, different scan parameters have been tested. The optimal settings for the confocal scan were $px_{\text{PSF}} = 100 \text{ nm}$, $\text{ROI} \approx 1.2 \times 1.2 \mu\text{m}^2$ and $t_{\text{exposure}} = 2000 \mu\text{s}$.

Following, the acquisition of the TCSPC data starts. Approximately every 6 ns, a pulse excites the sample with a different doughnut-shaped PSF. At the end of the measurement, the drift correction and TCSPC data are saved for the subsequent MINFLUX localization analysis.

3.4. Data Analysis Workflow

The acquired data measuring the drift in xyz , depicting the four doughnut PSFs and telling the relative and absolute photon arrival times require structured data analysis to estimate the emitter location. An existing Python script provided a basic workflow for the evaluation of those data.

The images from the PSF measurement are analysed to obtain the detection probabilities $p_i(\vec{r}_m)$. The images are interpolated to 1 nm pixel size and shifted compensating to the measured drift in x and y . The PSF images from the same doughnut excitation beam are summed. The four resulting images are cropped around the centre of PSF 1 leaving a FOV_{PSF} of 200-300 nm. The cropped PSFs are fitted by a 2D-polynomial using least-squares fitting employing the Levenberg-Marquardt algorithm. The fitted polynomials are evaluated

on a grid of 1 nm pixel size. Normalizing the four fitted images provides the $\lambda_i(\bar{r}_m)$ and further calculation according to equation 2.4.4 yields the $p_i(\bar{r}_m)$.

From a histogram of the absolute photon arrival times, the timing of the four excitation beam pulses is derived. A time window is selected for the MINFLUX estimation. The chosen time window is binned into sub-intervals of 0.1 s length. The relative photon arrival times per interval are evaluated for the events which were detected 0-2 ns after the four excitation pulses. Setting those four time windows denoted by $[t_{\text{init}}, t_{\text{crop}})$ prevents crosstalk between the interleaved pulses [92]. The sum of photon detection events counted within each of the four nanosecond time windows represent the n_i . Figure 3.6 visualizes the concept for preventing crosstalk contributions.

The calculated p_i and the n_i corresponding to the binned intervals are processed employing the MINFLUX MLE to find a position estimate for each interval. The estimated emitter locations are corrected using the drift data from the MINFLUX measurement. Statistical analysis of the estimated positions allows evaluating the achieved localization precision. The mean of the computed positions in x and y estimates the true emitter location. Using the fitted PSF images, the CRB is calculated for the estimated emitter position.

4. Optimization of Doughnut-Shaped PSFs in MINFLUX

This chapter studies the doughnut-shaped excitations PSFs focussing on how their measurement and fit influences the MINFLUX localization precision. A theoretical model for the PSF measurement is developed. The model is used in Monte-Carlo simulations evaluating the experimental parameters of the confocal scan of the PSFs. The power of GPUs is explored with the aim to accelerate the Monte-Carlo analyses. The optimal set of parameters for a PSF measurement is investigated. Finally, we propose a protocol to facilitate the alignment process of the MINFLUX EBP.

4.1. Modelling of the PSF Measurement

Taking advantage of the fact that imaging and fitting problems are well-described by known physical models, it is feasible to abstract and simulate the PSF and MINFLUX measurement. To assure a statically significant estimation of the variables of interest, we employ Monte-Carlo simulations. This widely-applied numerical method allows estimating the probability distribution of random and deterministic variables [93].

For the MINFLUX estimation, only the excitation intensities within the EBP triangle are of interest. This area displays the central part of the doughnut PSFs which is approximately a 2D-polynomial. The coefficients c_{ij} , $i, j \in \{0, 1, 2, 3, 4\}$ define the polynomial

$$I(x, y) = \sum_{i,j} c_{ij} (x - x_0)^i (y - y_0)^j, \quad (4.1.1)$$

with x_0 and y_0 allowing to shift the polynomial, e.g. to the desired EBP position. Throughout our study, we use coefficients $c_{02} = c_{20} = 10^{-2}$ and $c_{ij} = 0$ for all other indices i, j .

To ensure the good fit of all four doughnut PSFs, we choose a FOV_{PSF} of 250 nm with PSF 1 in its centre. The PSFs have a pixel size px_{PSF} modelling the confocal scan. A total number of N_{PSF} photons is distributed per imaged PSF assuming a multinomial distribution conditioned by the shape $I(x, y)$ of the doughnut PSF. For a given N_{PSF} , the respective signal-to-noise ratio (SNR) will increase with growing pixel size px_{PSF} . We choose to fix the total number of photons instead of the SNR because the absolute 'information' content of an image is carried by the detected photons. Experimentally, more photons per pixel can be collected by either reducing the speed of the confocal scan or by employing a different fluorescent species, e.g. gold nanoparticles (Au NPs) instead of fluorescent molecules.

We model a contribution of $N_{\text{PSF},b}$ background photons defined by the number of signal photons $N_{\text{PSF},s}$ and the SBR in the PSF measurement

$$N_{\text{PSF},b} = \frac{N_{\text{PSF},s}}{SBR_{\text{PSF}} + 1}. \quad (4.1.2)$$

The background photons are distributed amongst the pixels according to a Gaussian distribution with expectation value $\mu = N_{\text{PSF},b}/(\#pixel)$ and standard deviation b . Signal and background photons sum up to the total number of photons N_{PSF} . The created PSFs are fitted to a 1 nm grid again using the 2D-polynomial. We employ least-squares fitting using the Levenberg-Marquardt algorithm. Figure A.2 in the appendix shows exemplary doughnut-shaped PSFs and their corresponding fits.

The first part of this study focusses on evaluating the fit precision. As Thompson *et al.*, we evaluate the precision of the doughnuts' centre position which we define as the minimum of the fitted PSF. The results of n experiments are expressed in terms of the root mean square of the error Δx

$$\Delta x_{\text{rms}} = \sqrt{\frac{\sum_{i=1}^n \Delta x_i^2}{n}} = \sqrt{\frac{\sum_{i=1}^n (x_i - x_{\min})^2}{n}}, \quad (4.1.3)$$

with x_i the estimated and $x_{\min}$ the true x -coordinate of the zero. We only include the coordinates of the zero of PSF 1 in this analysis.

The simulation of the MINFLUX experiment requires the fit of four PSFs forming the EBP. The fitted and normalized PSFs define the probabilities $p_i(\bar{r}_m)$ where $\bar{r}_m$ states the emitter position. The photon number n_i corresponding to each excitation is obtained setting a total number of photons N_{MINFLUX} collected per localization. The number of signal photons N_s is derived using the relations

$$N_b = \frac{N_{\text{MINFLUX}}}{SBR_{\text{MINFLUX}} + 1}, \quad (4.1.4)$$

$$N_s = N_{\text{MINFLUX}} - N_b. \quad (4.1.5)$$

Here, SBR_{MINFLUX} is the SBR of the MINFLUX measurement.

The detected photon number n_i for each excitation is computed assuming a multinomial distribution for a fixed number of signal photons N_s and the probabilities p_i . Background photons N_b are distributed over the n_i according to a uniform distribution centred at $N_b/(\#pixel)$. Knowing p_i and n_i , the MINFLUX estimation is performed and the calculated localization accuracy and precision are returned. Both quantities are calculated as the average of x - and y -dimension. In the case of the localization accuracy, the corresponding mean 1D error is

$$\sigma_{1D} = \sqrt{\frac{1}{2n} \sum_{i=1}^n \left((x_i - x_0)^2 + (y_i - y_0)^2 \right)}, \quad (4.1.6)$$

where the x_i, y_i are the estimated emitter coordinates and x_0, y_0 , being the true emitter coordinates.

For the MINFLUX and PSF measurement, a Monte-Carlo simulation is realized repeating the previously outlined steps n times. All parameters remain the same for each of the n runs. The created 'experimental' PSFs will have slightly different photons distributions due to the simulated shot and background noise contributions. Figure 4.1 (a) sketches the outlined simulation workflow.

The simulation results will be compared to the CRB computed for the respective set of PSFs without noise. An exemplary PSF without noise is depicted in figure A.2 (d). As the fitted PSFs, the noise-free has $FOV_{\text{PSF}} = 250$ nm and $px_{\text{PSF}} = 1$ nm. The coefficients of the 2D-polynomial are the same. To prevent confusion with the simulated 'experimental' PSF, noise-free PSFs are normalized to one.

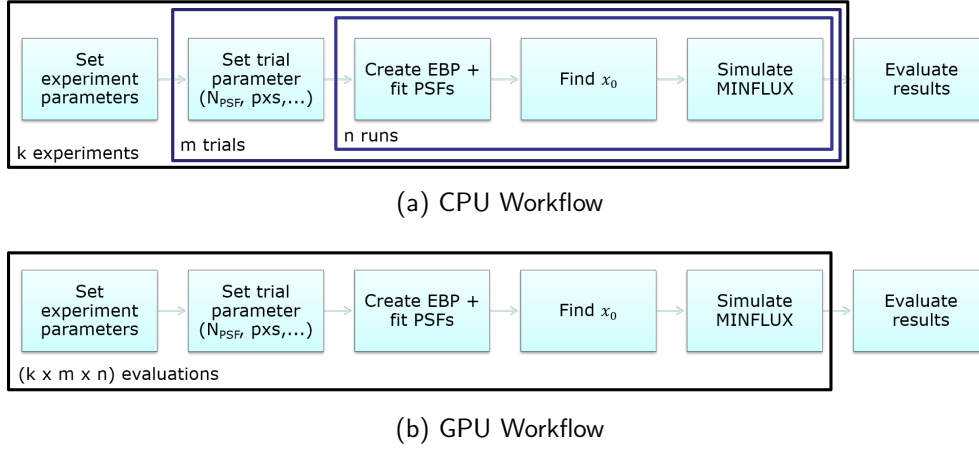


Figure 4.1.: Simulation workflows for Monte-Carlo analysis. (a) Employing the CPU, the code loops n times for the same set of parameters, returning the result and consequently jumping to the next trial changing parameters as N_{PSF} . After m trials, the next iteration of the k experiments is executed. From k experiment results, the standard deviation of the estimated parameter is derived. (b) Reducing data transfer between CPU and GPU all parameters for $(k \times m \times n)$ evaluations are copied at once to the GPU. The variable x_0 denotes the x-coordinate of the doughnut zero investigated to estimate the precision of the fit.

Besides comparing to the CRB, we moreover compare the simulated 'experimental' data to the result of a MINFLUX simulation using noise-free PSFs. This facilitates to evaluate the influence of the PSF measurement on the MINFLUX estimation. Comparing the simulation outcome for noise-free PSFs and the corresponding CRB will show the behaviour of the MINFLUX MLE compared to the theoretical lower limit expressed by the CRB.

4.1.1. GPU Computing

A common problem of Monte-Carlo simulations is the scaling of the error of the estimated quantity with $\frac{1}{\sqrt{n}}$. To gain one order of magnitude in precision is computationally expensive. However, Monte-Carlo simulations are easy to parallelize and open the door for multithreading on CPUs and even faster GPUs [93, 94].

By definition, the graphics processing unit (GPU) names the central chip of a graphics card. The name is often interchangeably used when referring to the graphics card itself as it will be done in this text. The GPU provides a visual output to the computer screen. The challenging demands of the computer games community during the last decades pushed the development of smaller and higher parallelized GPU architectures. State of the art gaming GPUs compute in the TFLOPS range whereas the latest CPUs provide up to hundred GFLOPS, making them at least one order of magnitude slower. An image or a video displays on a screen is a large array of structured data processed by the GPU. The logical conclusion is that GPUs can be employed for the execution of other tasks than only providing a video signal. Without the application of powerful GPUs, machine learning and artificial intelligence would not be discussed by the general public [95, 96].

In practice, running code on a GPU requires to partly rewrite it. The market leader for GPUs, Nvidia provides for their graphics cards the CUDA framework. CUDA enables to

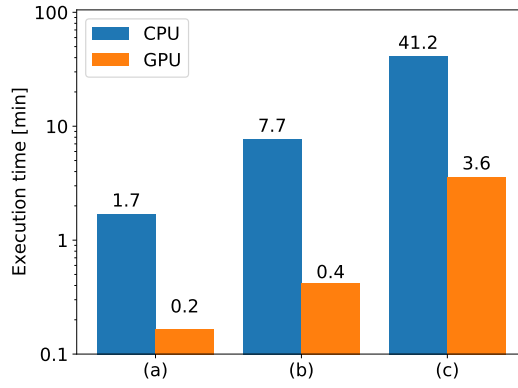


Figure 4.2.: Execution times for Monte-Carlo simulations on CPU and GPU. (a) Fit of Gaussian emission PSFs: fit of 130000 Gaussians on a (13×13) pixel grid. (b) Fit of doughnut PSFs: 26000 fits of a 2D-polynomial on a grid with (25×25) pixels. (c) MINFLUX estimation: 26000 MINFLUX estimations based on 104000 polynomial fits on a (25×25) pixel grid.

employ GPUs for parallelized computing [97]. High-level languages as Python or Matlab need a wrapper to interact with CUDA and thus the GPU. For general tasks, CuPy offers a great image of Python’s Numpy [98]. For more specific tasks, the PyCUDA and Numba packages offer to compile Python code to CUDA [99, 100].

GPU computing is especially interesting for speeding-up the computationally demanding fit of the doughnut PSFs [101]. For this task Przybylski *et al.*, developed the *Gpufit* package for least-squares fitting based on the Levenberg-Marquardt algorithm [102]. In this work, a short CUDA-script was developed. As an extension to Gpufit, this allowed fitting the doughnut PSFs on the GPU. The CUDA C++ script is shown in the code section of the appendix (see A.3.2).

The GPU is only efficiently employed when the data transfer between CPU and GPU is minimized and optimally all computations are performed in parallel. Running code with many loops will lead to continuous copying back and forth and ultimately be slower than only using a CPU. The simulation workflow was rewritten for parallelized processing (*cf.* figure 4.1). In figure 4.2, the gained speed-up is illustrated comparing the execution times for running three different simulations on CPU and GPU. The hard- and software configuration employed for the simulations is listed in table A.5 in the appendix.

In 4.2 (a), we simulated the fit of a Gaussian function to a simulated single emitter PSF. The fitted FOV_{PSF} was 13 px employing a Gaussian PSF centred in the FOV with $\sigma_{\text{PSF}} = 5$ px. From 130000 fits, we evaluated the fitted emitter position. The speed-up in execution time was 8.5-fold comparing CPU to GPU performance. A second test evaluated the fit of the 2D-polynomial describing the centre of a doughnut excitation beam. We set $FOV_{\text{PSF}} = 250$ nm and $px_{\text{PSF}} = 10$ nm. 26000 fits of the polynomial were performed. The precision of the fitted doughnut centre positions was computed. A 20-fold speed-up was achieved employing Gpufit.

When fitting the doughnut PSFs for the same parameters and consequently simulating a MINFLUX experiment, a speed-up of about $11\times$ was found computing on a GPU. This evaluation included 26000 estimations of the emitter position. Hence, 130000 doughnut polynomials were fitted. The speed-up was reduced compared to the sole fitting of the

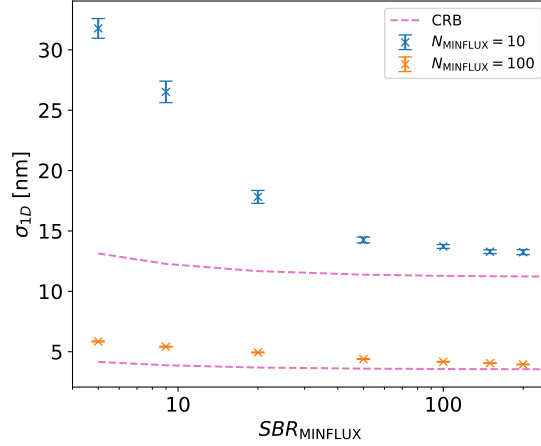


Figure 4.3.: MINFLUX localization precision as function of SBR_{MINFLUX} . For each data point, the simulation was run 5000 times. For the simulation, noise-free PSFs and an EBP with $L = 100$ nm were used.

doughnut PSFs as the simulation of the MINFLUX experiment could only partly be outsourced to the GPU. It was necessary to sequentially transfer data from the GPU to the CPU which reduced the overall speed-up. For future revisions of the code, it is recommended to employ PyCUDA or to write a complete C++ implementation for Nvidia’s CUDA. This will circumvent problems with cupy’s implementation of the numpy.random package.

4.2. Effect of the PSF Measurement on the Localization Precision

As a first test of the developed simulation workflow, we investigated the dependence of the MINFLUX localization precision on the signal-to-background ratio in the MINFLUX measurement. Noise-free PSFs and an EBP of $L = 100$ nm were used. The emitter was placed in the centre of the EBP. The simulation was repeated for 5000 runs for $N_{\text{MINFLUX}} = 10$ and $N_{\text{MINFLUX}} = 100$, respectively.

Figure 4.3 depicts the simulation results showing the estimated localization precision and the CRB for each N_{MINFLUX} . The localization precision improves with growing SBR_{MINFLUX} . For $N_{\text{MINFLUX}} = 100$, the precision converges to the CRB; for $N_{\text{MINFLUX}} = 10$, this is not the case. We show that the MLE only attains the CRB for large photon numbers wherein $N_{\text{MINFLUX}} = 100$ is sufficient for a high SBR_{MINFLUX} of 200. The simulation results agree with the analytically derived dependence on the SBR_{MINFLUX} (cf. eqn. S31 in [61]).

The employed noise-free PSFs replace an ideal PSF measurement where the result of the confocal scan and the subsequent fitting returns the true image of the doughnut excitation beams. The localization precision achieved for noise-free PSFs is the lower limit for a given set of parameters in the MINFLUX measurement. In contrast, the CRB evaluates the MINFLUX estimation assuming a perfect PSF measurement.

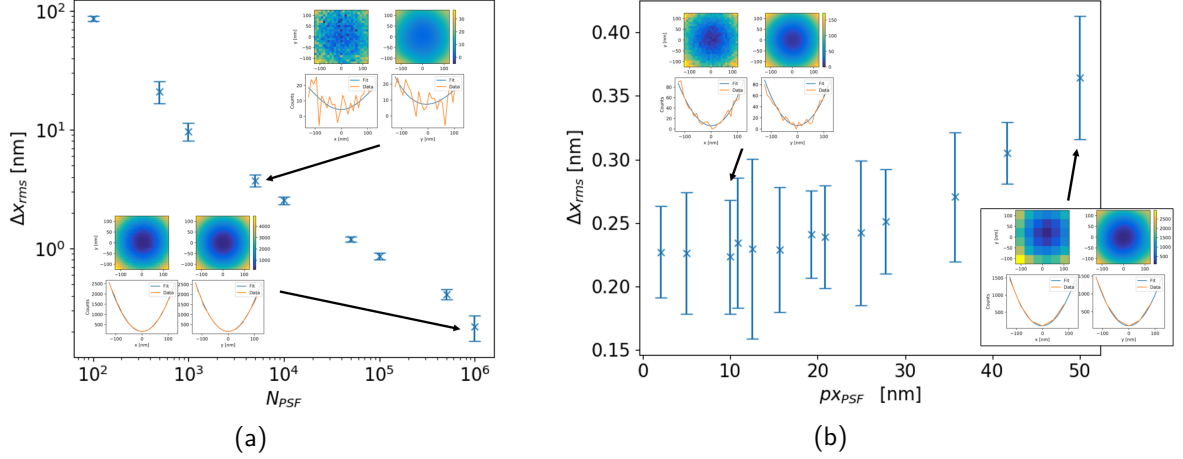


Figure 4.4.: Precision of fit of a doughnut-shaped PSF. Δx_{rms} evaluates the precision of the x-coordinate of the fitted doughnut zero. The PSFs were simulated and fitted using $FOV_{PSF} = 250$ nm. A Gaussian background contribution was modelled according to a SBR_{PSF} of 20 and $b = 3$. Figure (a) shows the dependence on the photon number N_{PSF} for $px_{PSF} = 10$ nm. In (b), the dependence on the pixel size px_{PSF} for $N_{PSF} = 10^6$ is depicted. Each data point is the result of 2000 simulations.

4.2.1. Fit of the Doughnut-Shaped PSF

Nanoscopy techniques based on centroid fitting are limited by the precision of the fit of the imaged emission PSF. Thompson and co-workers derived analytically the error of the fitted emitter position (*cf.* formula 2.4.11). To evaluate our model for the PSF measurement, we reproduced the results by Thompson *et al.* for the fit of a Gaussian emission PSF. Figure A.3 depicts the dependency of the estimated emitter position on (a) the collected photon number and (b) the pixel size of the camera detector. In both cases, the results agree perfectly with the theory.

The workflow for evaluating the centroid fitting was modified for the fit of the doughnut excitation PSFs. Similar to the evaluation of the x-error of the fitted centre position of the Gaussian, we calculate the error in the x-coordinate of the doughnut zero position. For every data point, the simulation was repeated for 2000 runs. The Gaussian background contribution was calculated setting the parameters $SBR_{PSF} = 20$ and $b = 3$. Both values are experimentally motivated (*cf.* figure A.1).

The simulation results in figure 4.4 show for the photon number dependence an improved fit precision with increasing photon number. Photon numbers on the order of thousands to millions are realistic as an excitation PSF is imaged. Sub-nanometre precision in the fitted zero position is one of the consequences of such high photon numbers. We find $\Delta x_{rms} \sim N_{PSF}^{-0.1}$ which proves a smaller dependence on the photon number than in centroid fitting. For $N_{PSF} > 10^5$, the error of the estimated precision grows which might be since the precision becomes smaller than the pixel size of the fitted PSFs.

The precision of the fit is worse for larger pixel sizes of the confocal scan. This is shown in figure 4.4 (b). Up to about 30 nm pixel size, the fit precision varies by less than 10 % compared to the precision at $px_{PSF} = 2$ nm. For larger pixel sizes, the error increases by 70 % at 50 nm pixel size. Compared to centroid fitting where the photon numbers are three

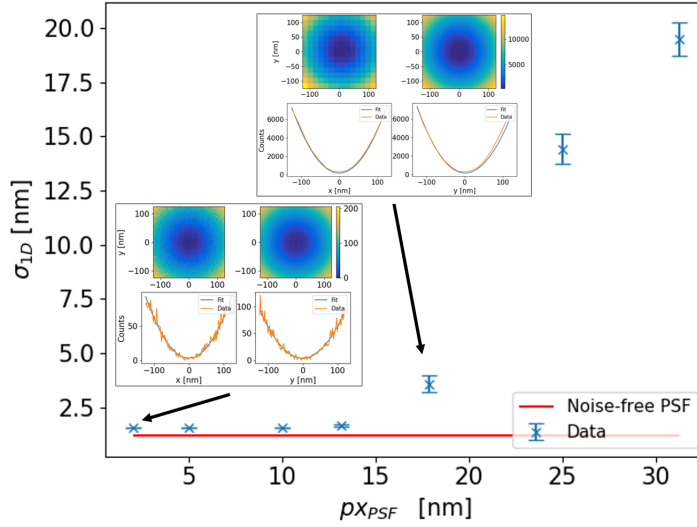


Figure 4.5.: Pixel size dependence of the localization precision. The simulation evaluates the attainable localization precision using different pixel sizes for the PSF measurement. Simulation parameter: $N_{\text{MINFLUX}} = 1000$, $SBR_{\text{MINFLUX}} = 20$, $N_{\text{PSF}} = 10^6$ and $SBR_{\text{PSF}} = 20$. Each data point results from evaluating 2000 simulations runs.

to four orders of magnitude smaller, background noise does not deteriorate the precision of the doughnut PSF fit when using small pixel sizes.

The estimation of the doughnut fit precision shows a similar proportionality on the photon number as in centroid fitting. The precision of the estimated doughnut zero position improves with the number of photons detected during the confocal scan. For the large number of collected photons during a PSF measurement, background noise does not have an influence on the fit precision and pixelation noise dominates.

Although the position of the doughnut zero is only one parameter describing the quality of the doughnut PSF fit, the findings indicate enhanced fit quality for PSF measurements with photon counts $> 10^4$ and pixel sizes < 30 nm. Subsequently, we investigate the influence of the choice of experimental parameters in the PSF measurement on the localization precision of a MINFLUX estimation.

4.2.2. Optimized Parameters for Robust PSF Measurements

To find the optimal experimental settings for a PSF measurement, we combine the code for the doughnut PSF fit and the MINFLUX estimation. Varying the parameters of the PSF measurement, we evaluate the change in localization precision of a MINFLUX experiment. The computed 'data' are compared to the localization precision achieved with noise-free PSFs.

Figure 4.5 illustrates the dependence of the localization precision on the pixel size of the confocal scan. Experimentally motivated, we set $N_{\text{PSF}} = 10^6$ and $SBR_{\text{PSF}} = 20$. The total number of photons N_{MINFLUX} emitted and evaluated in the MINFLUX measurement is 1000 with $SBR_{\text{MINFLUX}} = 20$. For the following evaluations, the emitter is positioned in the EBP centre. The localization precision was estimated after 2000 runs per parameter combination.

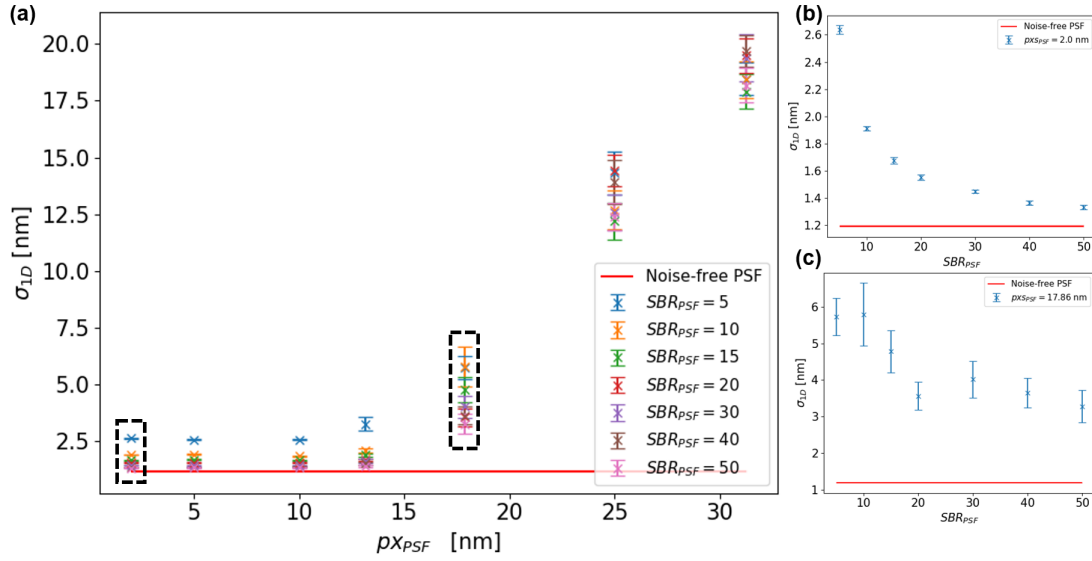


Figure 4.6.: Dependence of localization precision on pixel size and SBR_{PSF} . (a) Overview showing all evaluated pixel sizes and SBR_{PSF} . (b) SBR_{PSF} dependence for $px_{PSF} = 2$ nm. (c) SBR_{PSF} dependence for $px_{PSF} = 17.86$ nm. Remaining parameter as in fig. 4.5.

The simulation shows that the localization precision decreases with larger pixel sizes. For pixel sizes below 10 nm, the estimated localization precision converges to $\sigma_{1D} = 1.5$ nm. For noise-free PSFs, we estimated the localization precision to be $\sigma_{1D} = 1.195 \pm 0.005$ nm. For pixel sizes > 17 nm, the localization precision is significantly deteriorated.

Extending the analysis, the pixel size dependence is simulated for different SBR_{PSF} values. Figure 4.6 (a) presents an overview of the estimated localization precision for different pixel sizes and SBR_{PSF} . Subfigures (b) and (c) depict the gain in precision with increasing SBR_{PSF} for 2 nm and 17.86 nm pixel size. Higher SBR_{PSF} improve the localization precision for any pixel size. Only in the case of pixel sizes < 15 nm, the localization precision for noise-free PSFs is reached. The analysis reveals that higher signal-to-background ratios in the PSF measurement increase the MINFLUX localization precision. The limit for noise-free PSFs is attained for pixel sizes ≤ 10 nm requiring $SBR_{PSF} \leq 30$. The smaller the pixel size, the less the localization precision depends on SBR_{PSF} .

The previous studies were conducted for one fixed N_{PSF} . To conclude the study, we map five different photon numbers per doughnut PSF for five different SBR_{PSF} . In figure 4.7, the simulation results for 2000 runs per pixel are shown. The pixel size of the confocal scan was set to $px_{PSF} = 10$ nm and $SBR_{MINFLUX} = 20$ which are typical experimental values.

The map shows the expected increase in localization precision with higher SBR_{PSF} . The same dependence holds for the number of photons per PSF image. For $N_{PSF} \geq 10^5$, sub-2 nm precision is reached for $SBR_{PSF} \geq 20$. In figure A.4 in the appendix, three of the above maps are shown for different pixel sizes. For $px_{PSF} = 2.0$ nm, the precision depends only on SBR_{PSF} varying by about 1 nm for $5 \leq SBR_{PSF} \leq 30$. For $px_{PSF} = 15.38$ nm, SBR_{PSF} has no influence on the localization precision. Instead, the localization precision is determined by N_{PSF} improving more than 5-fold starting from $\sigma_{1D} = 27$ nm for $N_{PSF} = 10^4$ to $N_{PSF} = 10^6$.

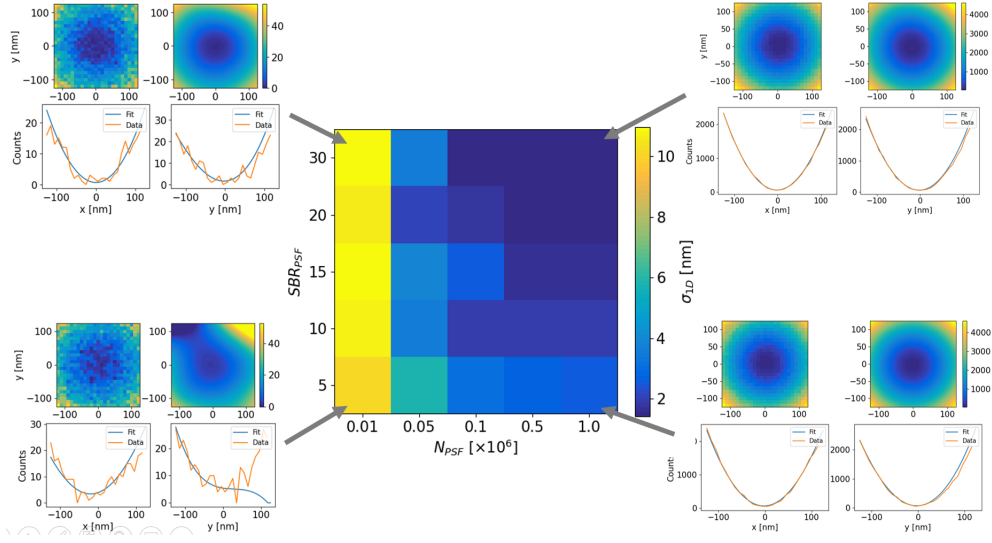


Figure 4.7.: Map of localization precision for different N_{PSF} and SBR_{PSF} combinations. The four insets show exemplary simulated and fitted PSFs for the four pixels at the corner of the map. For each pixel of the map, 2000 runs were carried out. The pixel size is $px_{\text{PSF}} = 10$ nm and $SBR_{\text{MINFLUX}} = 20$.

The last result indicates distinct ranges for the pixel size of the confocal scan where background and pixelation noise dominate respectively. Similar to centroid fitting, we find the background noise determining the localization precision in the case of small pixel sizes of 2 to 5 nm. For 10 nm pixel size, the influence of both noise sources is balanced. For $px_{\text{PSF}} \geq 15$ nm, pixelation noise dominates. In contrast to centroid fitting, the localization precision is less deteriorated by background noise than by pixelation noise because of the larger photon numbers N_{PSF} .

In summary, we demonstrated that the choice of parameter in the PSF measurement has a significant influence on the attainable localization precision in a MINFLUX experiment. The localization precision of noise-free PSFs is reached for $N_{\text{PSF}} \geq 10^5$, $SBR_{\text{PSF}} \geq 20$ and $px_{\text{PSF}} \leq 5$ nm. We showed that small pixel sizes lead to more robust localization precisions varying by less than 1 nm for $SBR_{\text{PSF}} \leq 30$.

In an experimental realization, the smallest possible pixel size for the confocal scan is limited by the resolution of the piezo stage moving the sample. The photon number acquired per pixel can be increased by reducing the speed of the confocal scan or by employing a different sample which emits more photons. The signal-to-background ratio depends on the design of the experimental setup and considerable on autofluorescence from the glass slide and coverslip.

It is recommended to use the smallest possible pixel size when imaging the doughnut excitation PSFs. Averaging several images per doughnut will smooth artefacts from, e.g. blinking events. The exposure time should be as high as possible to collect a maximum of photons. The use of a bright fluorescent species, for example, a metal nanoparticle or a bead, is advantageous compared to a single molecule as it will not bleach when illuminated with high laser powers or for long exposure times.

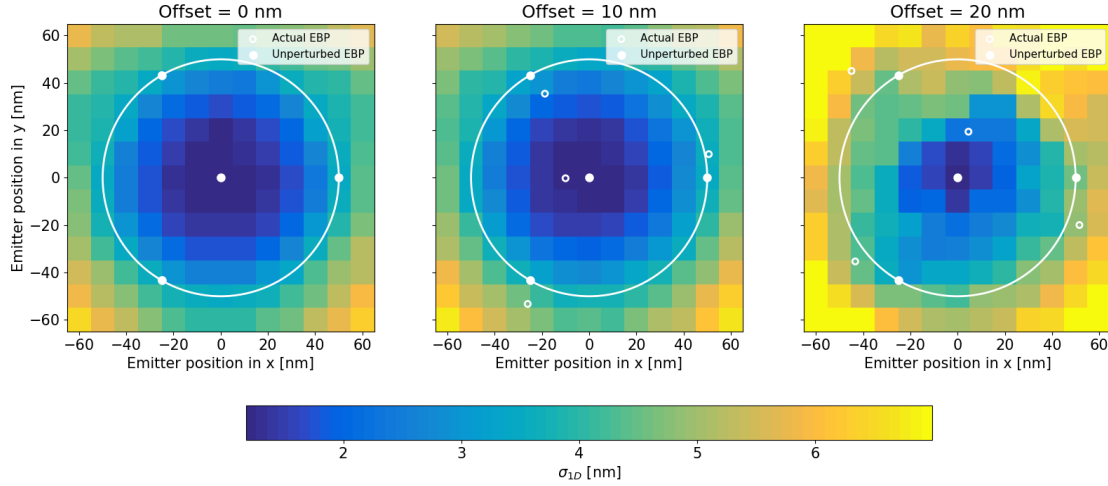


Figure 4.8.: Comparison of emitter maps for different EBP misalignments. Simulation for $SBR_{\text{MINFLUX}} = 20$, $N_{\text{MINFLUX}} = 1000$ and noise-free PSFs. 10 randomly chosen EBPs with fixed radial offset were evaluated; 100 times each.

4.2.3. Misalignment of Excitation Beam Pattern

To evaluate the localization precision at different emitter positions, we create a map by simulating a MINFLUX experiment with noise-free PSFs. Each pixel of this map is the result of 1000 simulation runs. We set $SBR_{\text{MINFLUX}} = 20$ and $N_{\text{MINFLUX}} = 1000$. Figure A.5 in the appendix reports the localization precision reached at the different emitter positions.

We have investigated the dependence of the MINFLUX estimation result on the PSF measurement assuming that the EBP is well-aligned. The following analysis considers a constant offset of the actual position of the four doughnut zeros to the desired zero positions of an unperturbed EBP. This models a statistical uncertainty in the accuracy of the EBP alignment process. Offset values from 0 to 25 nm are evaluated. The radial offset is fixed to the set value. The angular direction of the offset zero position is randomly chosen for each of the 4 excitation doughnuts. $SBR_{\text{MINFLUX}} = 20$ and $N_{\text{MINFLUX}} = 1000$ were used as experimental parameters. For every offset value, a map for different emitter positions was computed. We set 1000 runs per pixel of the map. After 100 runs, a new, randomly offset EBP was created. This ensured that a particular EBP giving by chance excellent localization precision did not determine the estimated value but remains an outlier.

In figure 4.8, three maps for three different offset values are shown. In the case of no offset, actual and unperturbed EBP are the same. For 10 nm and 20 nm offset, one of the ten random EBPs is overlayed to the computed map. We find that for 0 nm and 10 nm offset, the maps are approximately identical. For 20 nm offset, the localization precision is deteriorated and only for emitter positions within 10 nm around the EBP centre, sub-2 nm precision is provided.

For the evaluation of the localization precision for the different offset values, the estimated value $\sigma(0,0)$ for the emitter placed at the EBP centre and a mean precision $\langle\sigma\rangle_{\text{FOV}}$ over the mapped emitter position space are calculated. In figure 4.9, both quantities are plotted as a function of the radial EBP centre offset. The simulation shows that the localization precision for an emitter at the EBP centre only changes for an offset larger than 20 nm. The precision drops from $\sigma(0,0) = 1$ nm to $\sigma(0,0) = 8.5$ nm. Evaluating the mean localization

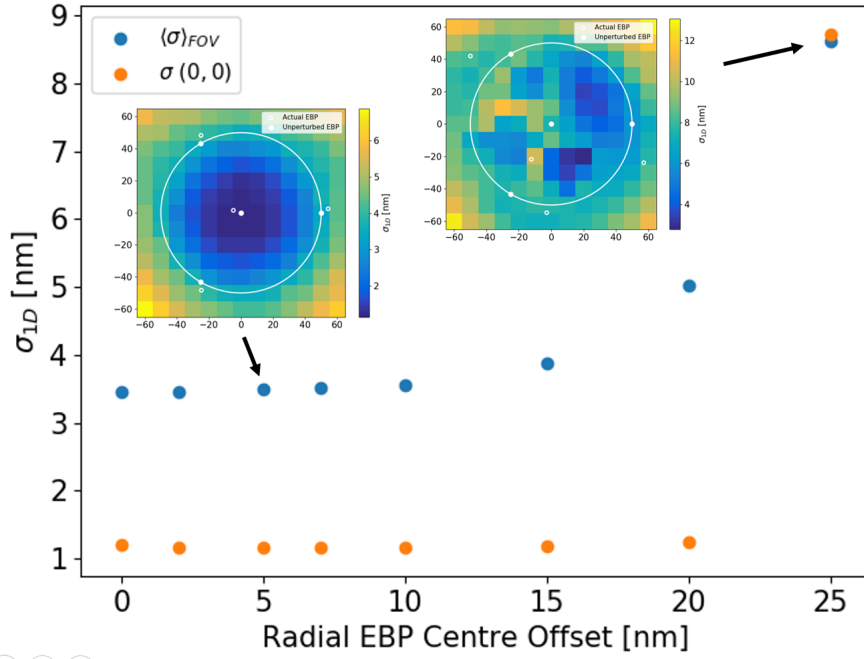


Figure 4.9.: Robustness to EBP misalignment. The localization precision evaluated at the EBP centre and the mean over the whole emitter position space are plotted for different radial EBP centre offset values. The simulation parameters are the same as in fig. 4.8.

precision, changes in $\langle \sigma \rangle_{FOV}$ of less than 0.5 nm are observed for offsets ≤ 15 nm. Larger offsets lead to decreased localization precisions.

We derive from this result that the attainable localization precision is robust against misalignments of the EBP. Perturbations of the doughnut zero position of less than 15 nm do not lead to a significant drop of the localization precision over the whole emitter position space. The zero of a doughnut excitation beam is well imaged within $(10-20)^2$ nm² assuming $px_{PSF} = 10$ nm. This value would be critical considering the 15 nm offset which we found to tolerate losses in localization precision. Smaller pixel sizes will circumvent this problem. A well-aligned doughnut excitation beam with a clearly-defined zero further facilitates the precise positioning of the doughnut zero to the desired EBP position.

4.3. Experimental Protocol for the Excitation Beam Alignment

In p-MINFLUX, four excitation beams have to be aligned to create four doughnut-shaped excitation PSFs matching the desired EBP arrangement in the focal plane of the objective. Subsequently, we enumerate and explain the steps carried out for the EBP alignment.

1. The four beams were aligned macroscopically using two alignment pinholes.
2. A fine-alignment was realized employing an auxiliary CMOS camera. The camera was shifted to two positions between L_1 and L_2 and the beams' spot position were adjusted to match.
3. A pinhole (P75D, Thorlabs Inc.) was inserted in the focal plane of L_1 to clean the

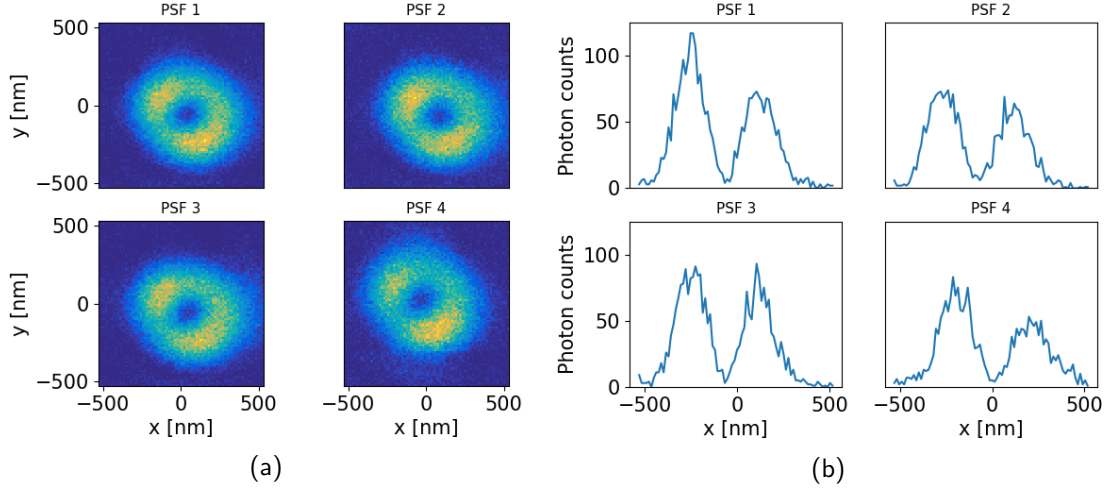


Figure 4.10.: Aligned doughnut-shaped excitation PSFs. (a) Cross-sections in xy . (b) Cross-sections through doughnut minimum. Confocal images acquired at a dwell time of 0.5 ms with 13 nm pixel size.

TEM-mode profiles and optimize the alignment. The alignment improved when the intensity measured behind the pinhole could be increased. Different pinhole sizes were tested. A diameter of 75 μm was found to provide a good compromise between a spatially-cleaned TEM_{00} mode profile and sufficient flexibility to align the EBP without cutting off any of the beams.

4. Without placing the VPP, the focal spot was checked to be diffraction-limited in all three dimensions. A dedicated PyFLUX function was implemented to perform a live fit of the imaged excitation PSFs with a Gaussian function. For fluorescence emission at $\lambda = 647 \text{ nm}$, the reference values are $\text{FWHM}_{x,y} = 231 \text{ nm}$ and $\text{FWHM}_z = 660 \text{ nm}$ where FWHM denotes the full width half maximum of the imaged emission PSF. Au NPs were illuminated to collect a strong and constant fluorescence signal. The excitation power was set to 10 μW .
5. The VPP was positioned in the conjugated plane to the objective's BFP. We experimentally matched the orientation of the VPP's spiral staircase to the orientation of the $\lambda/4$ -waveplate.
6. The shape of the four doughnuts was evaluated. Residual misalignments in the common path of all four beams were removed by beam-walking the tilt of the last two mirrors in front of the objective.
7. The alignment of the MINFLUX EBP was done locking the xyz-positions and scanning a FOV of $\sim 0.4 \mu\text{m}$ size. The FOV showed the doughnut centre of the four pre-aligned beams exciting a Au NP.
8. The four initially overlaying beams, were aligned to an EBP with $L = 100 \text{ nm}$. The doughnut zero positions were translated by minutely tilting the M_c mirrors. Because of the placement of the mirrors in the plane conjugated to the BFP of the objective, a tilt creates a lateral movement of the focus without misaligning the beam along the optical axis. Within PyFLUX, the four targeted zero positions of the EBP are projected on the live image of the confocal scan to facilitate the alignment process.

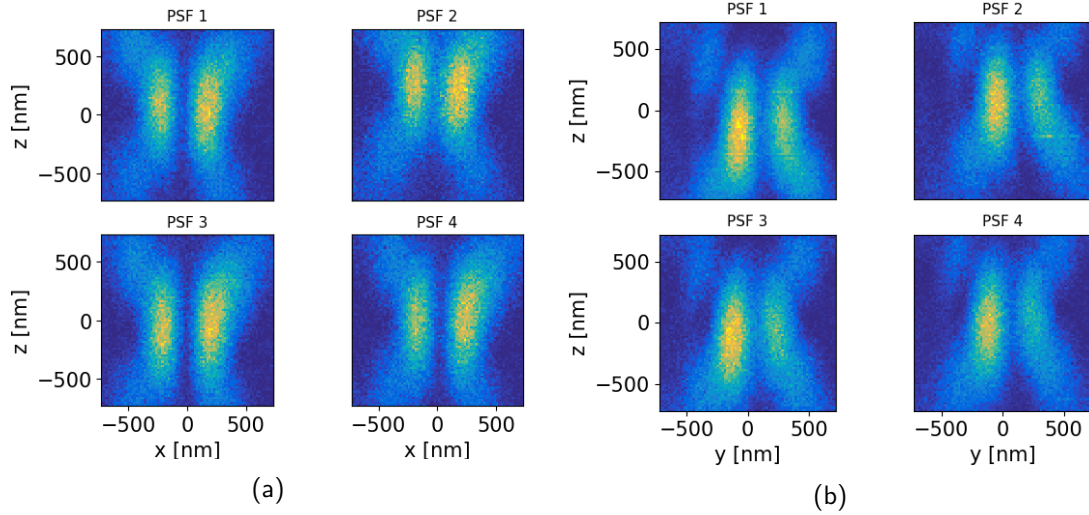


Figure 4.11.: Aligned doughnut-shaped excitation PSFs. (a) Cross-sections in xz. (b) Cross-sections in yz. Confocal images acquired at a dwell time of 0.5 ms with 18 nm pixel size.

Figure 4.10 (a) depicts a typical dataset of the aligned p-MINFLUX excitation doughnuts. The images were acquired while active drift correction was on. In contrast to the actual PSF measurement, we chose a scan area $> 1 \times 1 \mu\text{m}^2$ to better evaluate the PSF's shape. The xy-cross-sections show four very similar doughnuts. From the alignment process, we learned that the similarity was mainly due to the placement of the pinhole.

The cross-sections in 4.10 reveal that the four doughnuts do not possess a centre without photon counts. A stochastic background contribution is one explanation for the photon counts. Furthermore, any remaining misalignment will deteriorate the beam shape and the quality of the zero. PSF 3 shows outside the excited area fewer photon counts than in the centre. Also, the doughnut is symmetric contradicting both of the above two explanations. We resume that the imperfect zero exists because of the excitation with elliptical instead of circular polarized light [68, 67]. As documented in table A.2, polarization contrast for PSF 3 and 4 is about 10 % compared to 2.5 % for PSF 1. The quality of the doughnut zero will benefit from further optimization of the incoming polarization state.

To evaluate the excitation beam for a tilt, cross-sections along the z-axis were acquired. The images of this analysis are shown in figure 4.11. Except for PSF 2, the cross-sections in xz are equal. A small tilt is visible. In yz, the picture is different as none of the four doughnuts shows a clear area of zero intensity in the beam centre. A minimal tilt from the orthogonal exists. Prominent is the shape of the side lobes which are not symmetric in z. This asymmetry also exists in the cross-sections in xz why we conclude that those artefacts are due to the refraction of the emitted light from the Au NP at the sample-coverslip and coverslip-immersion oil interfaces [103].

The visible asymmetry and elongation in y might be proof of a misalignment of an optical component leading to a situation where the incident beam is slightly uncollimated in y when entering the BFP of the objective. This is, for example, possible when a dichroic mirror is modestly bent along one axis. Comparing the z positions of the regions of greatest intensity for the four PSF, we find a slight offset of the focal position of PSF 2.

We showed that it is possible to align four different beams independently to form similarly tailored doughnut PSFs. Employing a custom camera tool and introducing a pinhole facilitated the alignment process significantly. The presented doughnuts are not perfect. Improving the fine-alignment of the optics and optimizing the incident polarization state will provide symmetric doughnut PSFs exhibiting true zeros. As we show in the subsequent section, imperfect excitation PSFs do not prevent from reaching high localization precisions in a MINFLUX measurement. A prerequisite is a PSF measurement providing an accurate image of the doughnut PSFs.

5. Nanometric Localization Precision by p-MINFLUX

Following the optimization and alignment of the EBP, the localization precision and accuracy provided by p-MINFLUX was tested. The doughnut PSFs of the aligned EBP were imaged during a PSF measurement. Five images of each excitation PSF were acquired within a *FOV* of 0.55 μm , using $px_{\text{PSF}} = 6.9 \text{ nm}$ and 0.5 ms dwell time. An area of $300 \times 300 \text{ nm}^2$ around the zero of the central EBP doughnut was fitted for each of the four images.

We observed sudden blinking of the Au NPs when the nanoparticle was excited too intensively during the PSF measurement. This phenomenon has been reported in the literature although it is not fully understood and subject to current research [40, 104]. To prevent the blinking, the excitation laser power of the pulsed doughnut beams was set to 10 μW at the back focal plane of the objective. PSF images depicting strong blinking were discarded before the PSF fit. Any remaining artefacts were smoothed by averaging several images and the PSF fit.

The IR beam of the focus lock system is able to excite fluorescence of the Au NPs [105]. Precise alignment of the IR excitation to ensure $> 2 \mu\text{m}$ distance to the EBP circumvented this problem.

The setup control software PyFLUX displays live images of the PSF measurement. This allows directly evaluating the results, changing the parameters and rerunning the measurement if necessary. In the case of the MINFLUX measurement, the TCSPC data are displayed in PyFLUX but no further information on the measurement outcome is available. The data analysis must be performed off-line with dedicated scripts and algorithms.

5.1. Graphical User Interface for Data Analysis

To facilitate the data analysis and accelerate the evaluation of measurement results, a general user interface was developed. A step-by-step analysis pipeline allows adjusting the analysis parameters for the evaluation of PSF and MINFLUX measurements.

The GUI was programmed using the Python-binding of the cross-platform application development framework Qt and the Python package pyqtgraph [106, 107]. Designed in QtDesigner, the user interface is easily adjusted to the requirements of new users. Figure 5.1 shows a screenshot of the current version of the analysis GUI. At the moment, the software provides

- Visualized PSF fitting with the optional saving of the fit for later analysis
- Automatic drift correction for data from PSF measurements
- Manual selection of trace segments from the TCSPC recorder
- Automatic processing of traces exhibiting blinking events

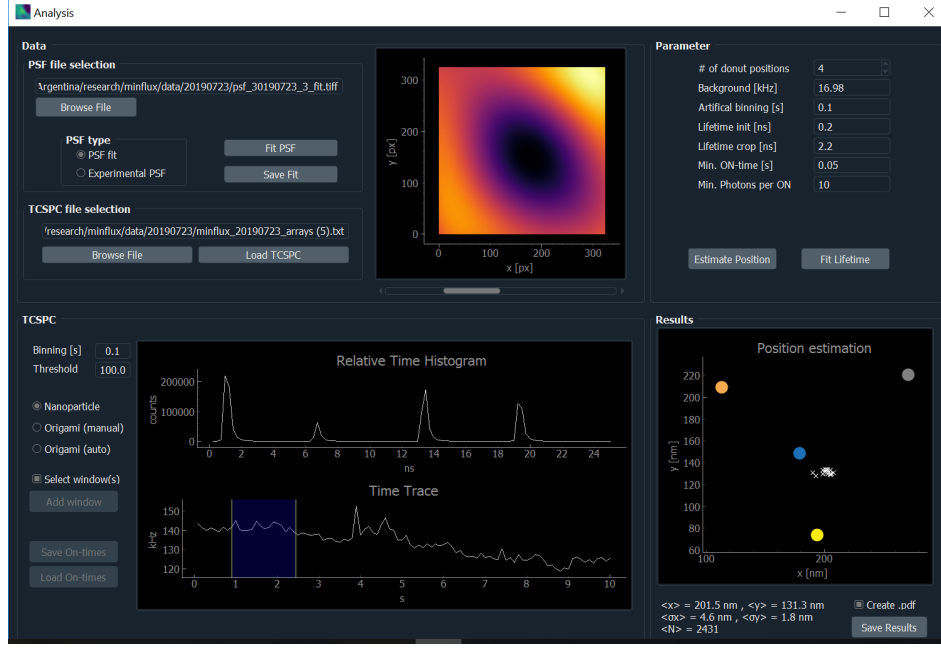


Figure 5.1.: Graphical user interface for data analysis. The GUI is structured by four modules for (1) the data selection and PSF fitting (top-left), (2) setting the desired analysis parameters (top-right), (3) the evaluation of the TCSPC data (bottom-left) and (4) the display of the analysis results (bottom-right). The GUI was tested with experimental data for the localization of Au NPs adapted from Luciano Masullo [23].

- Extraction of lifetime information
- Mapping of localized emitter positions.

The software allows to automatically calculate the signal-to-background ratio given a previously measured count rate for the background noise. The SBR_{MINFLUX} is computed for every time window selected in the GUI. Additionally, it is possible to select the nanosecond time windows $[t_{\text{init}}, t_{\text{crop}})$ in the histogram of the relative photon arrival times. The presented GUI was used for the analysis of the experimental measurement data from the laboratory.

5.2. Localization Precision vs. Photon Number

In a first step to validate p-MINFLUX, we localized gold nanoparticles. TCSPC data were acquired for 5 s. The measured absolute and relative photon arrival times are shown in the histograms in figure 5.2. In contrast to fluorescent molecules, Au NPs usually do not blink. As depicted in the time trace in figure 5.2 (a), the count rate is approximately constant. A trace from a single fluorescent molecule would be binned according to its on-times. The position of the molecule would be estimated for each of those bins.

Given a constant count rate of ~ 400 kHz, we bin the data to a fixed number of intervals. We aimed to analyse the localization precision depending on the total number of photons regarded for one position estimation. A custom-written Python algorithm calculates the adequate bin sizes. To avoid crosstalk, we choose time windows $[t_{\text{init}} = 0 \text{ ns}, t_{\text{crop}} = 2 \text{ ns})$

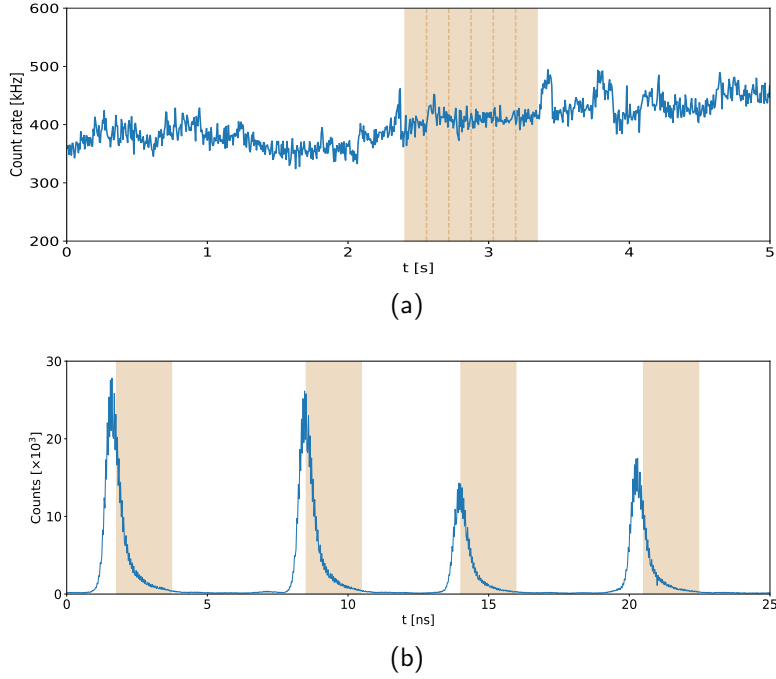


Figure 5.2.: Histograms of the absolute and relative photon arrival times. In (a), the fluorescence trace and the time window for the position estimation are depicted. The dashed lines mark the six binned intervals. Figure (b) shows the relative photon arrival times. The four excitation pulses were detected at $t \approx 1.7, 8.5, 14.0,$ and 20.5 ns. The coloured intervals indicate the 2 ns-long time windows for the selection of the detection events.

for the relative photon arrival times. The histogram of the relative photon arrival times in figure 5.2 (b) depicts those time windows. We find the relative arrival times of the four pulses at $t \approx 1.7, 8.5, 14.0,$ and 20.5 ns.

All four doughnuts beams had the same excitation intensity of $10 \mu\text{W}$ at the objective's BFP. Hence, from the counts of each of the four peaks relative photon arrival times, we expect the emitter to be positioned close to the centre of doughnuts 3 and 4. An independent background measurement found a background count rate of ~ 17 kHz (*cf.* table A.4). For the selected part of the fluorescence trace, the corresponding SBR_{MINFLUX} is 24.

Using the fitted PSFs and the selected photon count data, the maximum likelihood estimation is carried out. After binning, each window contained $N = 5000$ photons regarded for the position estimation. Here and throughout this chapter, we drop the MINFLUX -subscript for the photon number detected in the MINFLUX measurement. An exemplary map of the resulting log-likelihood l is shown in figure 5.3 (a). The map is the result of the MINFLUX estimation for one time window. It displays the likelihood of a position (x, y) that the emitter was positioned at this exact location during the measurement. The maximum in the log-likelihood map is marked by a red star. The position of the centre of the three outer doughnuts corresponds to the areas with the lowest likelihood values. In agreement with the relative photon arrival histogram, the emitter was not positioned at any of the zero positions as none of the four peaks vanished.

Repeating the sketched estimation for the $M = 35$ binned windows, we obtain 35 estimated

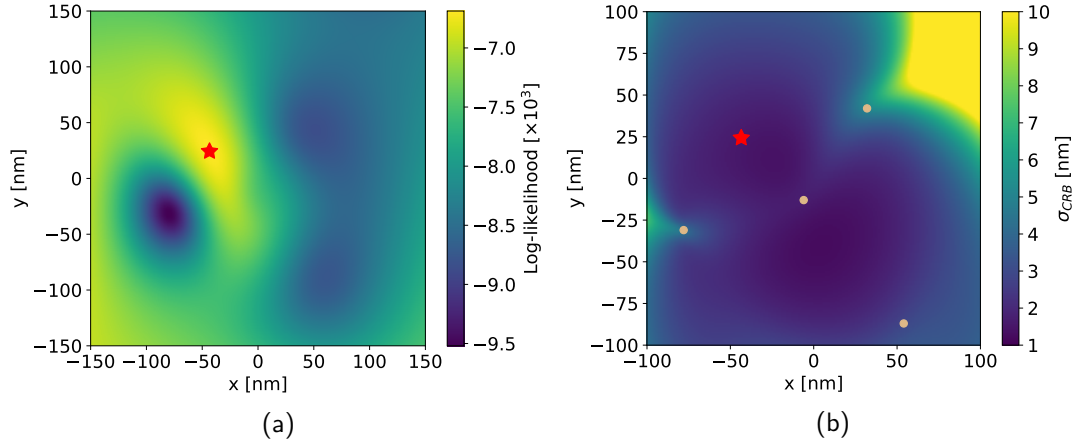


Figure 5.3.: Comparison of experimentally determined log-likelihood and theoretical CRB. One time window with $N = 5000$ photons was evaluated. In the log-likelihood map in (a), the centres of the three peripheral doughnuts are recognized as areas of smallest likelihood. In the map of the CRB in (b), dots indicate the four centre positions of the doughnut-shaped excitation PSFs. $L \approx 100$ nm. A red star marks the estimated location of the emitter.

emitter positions. Figure 5.4 depicts a histogram of the estimated emitter positions. The standard deviation of the estimated x- and y-positions was computed to find $\sigma_x = 1.5$ nm and $\sigma_y = 1.4$ nm. The corresponding, mean 1D localization precision is $\sigma_{1D} = \frac{\sigma_x + \sigma_y}{2} \sim 1$ nm. We thereby prove that p-MINFLUX is capable of localizing with 1 nm precision.

We compare the experimentally achieved localization precision to the CRB for the given set of excitation PSFs. In figure 5.3 (b), the CRB is mapped for the given doughnut PSFs. Additionally, the doughnut zero positions and the estimated emitter position is shown. As expected, we find the emitter localized between the centres of PSFs 3 and 4.

The outlined analysis for $N = 5000$ was repeated for different photon numbers. The localization precision for each N and the CRB for a MINFLUX estimation are visualized in figure 5.5. The error bars for the experimental localization precision were calculated by a bootstrapping analysis [108]. The measured localization precision attains for $N > 500$ the CRB. As expected for the MLE, the localization precision does not reach the CRB for small photon numbers. When comparing our experimental results to the CRB of a camera-based localization scheme, the superiority of MINFLUX becomes clear. $5\times$ fewer photons are required to reach 5 nm localization precision compared to a perfect camera-based detector with infinite SBR . For a camera-based localization technique and the same signal-to-background ratio as in our measurement, *i.e.* $SBR = 24$, $100\times$ more photons are necessary to attain the 5 nm precision.

For small N , the histograms in figure 5.5 (b) show a spread of the localizations along a preferential axis. This result is expected since MINFLUX does not provide isotropic precision in x and y. We assume that the precision decreases more rapidly along $y=-x$ as the modulation of PSF 3 and 4 remains equal (*cf.* fig. 5.3 (b)). As expected from the theory, we observe that MINFLUX delivers better localization precision in regions of stronger modulation of the excitation. Refinement of the pre-positioning procedure in PyFLUX will allow placing the

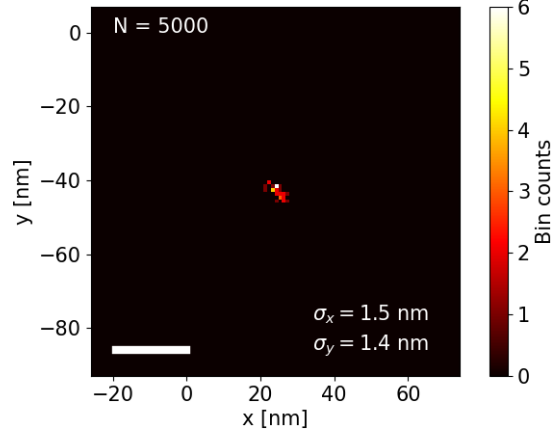


Figure 5.4.: 2D-histogram of estimated emitter positions. $M = 35$ time windows with each $N = 5000$ photons were evaluated. A gold nanoparticle was localized with a precision of ~ 1 nm. The estimated position corresponds to the location of maximum likelihood indicated in figure 5.3.

emitter closer to the centre of the EBP. We expect to achieve 1 nm precision for less than the reported $N = 5000$ photons.

5.3. Demonstration of Localization Accuracy

After testing the achievable localization precision in p-MINFLUX, we studied the localization accuracy. A first toy model saw a controlled stage movement during an active MINFLUX measurement. Although the Au NPs are 60 nm in size, their centre is localized with up to 1 nm precision allowing to image shifts in the particle position smaller than the particle diameter.

We moved the stage twice along the x-axis. Each translation was set to a distance of $d_x = 20$ nm. Three 2 s-long time windows were analysed. We binned the TCSPC data for a binwidth of 0.05 s and $[t_{\text{init}} = 0 \text{ ns}, t_{\text{crop}} = 2 \text{ ns})$. On average, 3000 were regarded for the MINFLUX estimation. The mean SBR_{MINFLUX} was 20. Apart from the active drift correction during the measurement, the estimated emitter positions were post-processed correcting for the measured drift in x and y. In figure 5.6 (a), the localizations of the three Au NP positions are shown. The three positions are clearly resolved. The shift of the position is mainly along the x-axis but also in y which might be due to the reduced localization precision in y or deviations between set and executed stage movement.

As shown in figure 5.6 (b), the measured localization precision in x is $\sigma_x = 2.3, 2.8$ and 2.9 nm where we enumerate for growing x-position. The two calculated distances between the estimated emitter positions agree well with the set 20 nm. This result shows that p-MINFLUX clearly resolves a 20 nm translation of the sample with precisions of 2 nm which correspond to approximately 4-5 nm spatial resolution.

We demonstrated a proof of concept of p-MINFLUX. Four laser beams were aligned to create four identical doughnut-shaped PSFs. 1 nm precision was achieved localizing a Au NP with the optimized EBP. Predicted by the theory, we experimentally confirmed that the

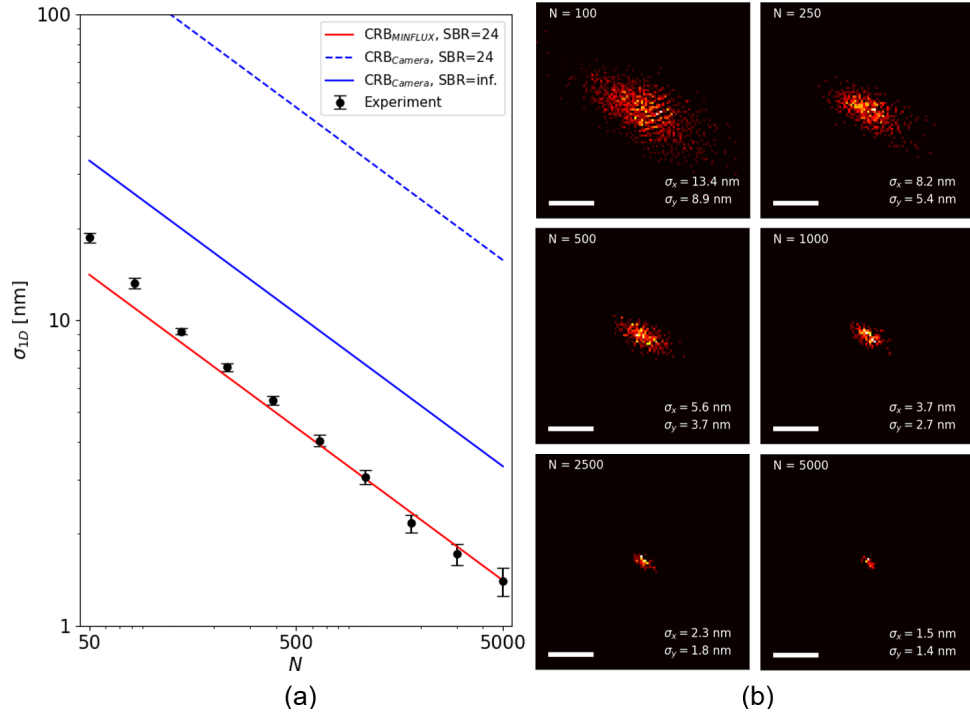


Figure 5.5.: Dependency of p-MINFLUX localization precision on photon number. Figure (a) compares the experimental and theoretical localization precision for different N values. The experimental values attain the CRB for $N > 500$. The CRB for a camera-based localization with $SBR = 24$ and infinite SBR are plotted. The parameter for the camera detector are the same as in figure 2.8. p-MINFLUX provides superior localization precision at the same N compared to any camera-based technique. In (b), 2D-histograms show the anisotropic spread of the estimated emitter positions with decreasing photon number N . The scale bars in (b) measure 20 nm.

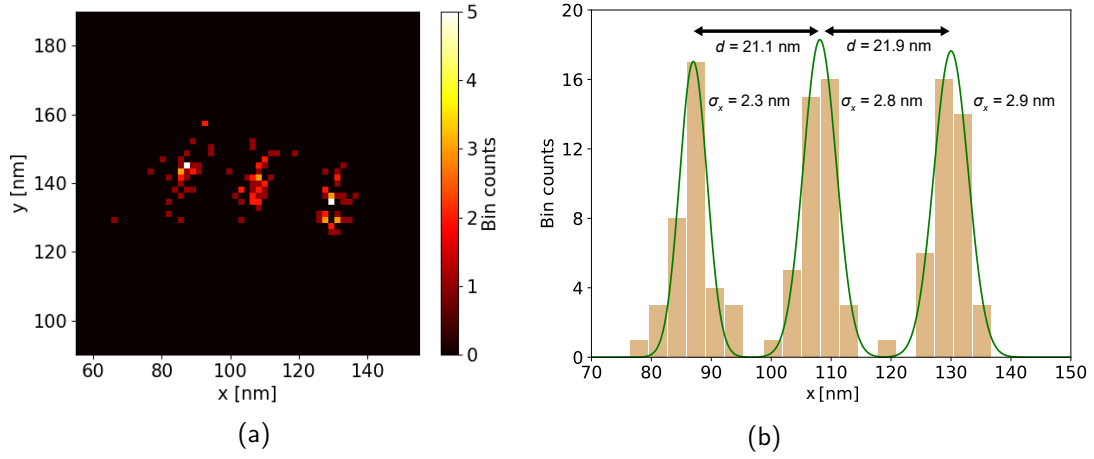


Figure 5.6.: Imaging of controlled Au NP movement. The sample was moved translating the stage by $2 \times d_x = 20$ nm. Three 2 s-long time windows with $\langle N \rangle = 3000$ were evaluated. The mean SBR_{MINFLUX} was 20. In (a), the estimated emitter positions are visualized in a 2D-histogram. Figure (b) depicts a projection of the estimated nanoparticle localizations on the y-axis.

MINFLUX MLE attains the CRB. The results indicate that p-MINFLUX provides more than 20 nm spatial resolution. For a full validation of p-MINFLUX, single-molecules need to be localized to directly compare to the localization precision and spatial resolution proven for the original MINFLUX realization. EBPs with different diameters should be tested to prove the linear proportionality of the localization precision on L .

The pinhole introduced to the experimental setup currently allows the alignment of EBPs with $L < 200$ nm. For larger L , the outer beams will be clipped by the pinhole aperture. For the EBP with $L = 100$ nm, a pinhole of 50 μm diameter would further facilitate the alignment and enhance the similarity of the four doughnut PSFs. The optimization of the doughnut PSFs could not fully remove a persisting astigmatism in the optical system. The larger elongation of the doughnut PSFs along y might be caused by the unwanted curvature of an optical element.

Removing the astigmatism will reduce the FWHM_y of the excitation PSFs, reduce the peak-to-peak distance of the doughnut PSFs and increase the modulation around the zero. One reason which prevents a true zero at the doughnut's centres is the polarization state of the light entering the objective. In the current setup, the axes of linear polarization are changed turning the fiber couplers. Introducing four $\lambda/2$ -waveplates behind each of the fibers would allow aligning the four polarization axes more precisely.

It was shown that the Au NP were illuminated with the EBP centre being significantly offset to the emitter. This reduced the attainable localization precision. The positioning of the piezo stage at the begin of the MINFLUX measurement will be improved by selecting a slower speed for the confocal scan which reduces the offset between forward and backward image. Additionally, the PyFLUX routine computing the actual emitter position from the two images needs to be refined. We tested the achievable resolution of the nanoscope translating the piezo stage along the y-axis. To provide a clearer picture for conclusions on the performance in 2D, predefined patterns scanning along both axes should be implemented.

6. Conclusion and Outlook

The position estimation in MINFLUX nanoscopy relies on precise knowledge of the employed excitation pattern. This work studied the characterization of the illumination arrangement by confocal imaging and its influence on the attainable localization precision.

Monte-Carlo simulations were conducted to find the optimal parameters for the measurement of the excitation pattern. A GPU-implementation of the Monte-Carlo algorithms was developed to access more computational power and speed-up the analysis. Optimum conditions for the characterization are obtained when (a) the number of detected photons is maximum, and (b) the smallest possible pixel size is used for the scan. More collected photons increase the signal-to-noise ratio and result in a more precise image. The second condition arises from the fact that the contribution of background noise in the measurement of the excitation pattern is negligible. Another set of simulations was performed to address the tolerance to misalignments of the excitation pattern. The analysis exhibited that the MINFLUX estimation shows negligible deviations (< 0.5 nm) for misalignments of up to 15 nm.

The simulations disregarded blinking or bleaching of the localized emitter. Those time characteristics can be included in future studies. This will also allow estimating the temporal resolution of p-MINFLUX and provide a comprehensive simulation workflow. Our study evaluated one single emitter. Simulating many emitters would enable the estimation of the achievable spatial resolution. Furthermore, the MINFLUX estimators could be compared for their performance when providing structural information [74].

A protocol for the alignment of four individual doughnut-shaped PSFs was established in the laboratory. The position measurement of a gold nanoparticle demonstrated 1 nm localization precision reached by p-MINFLUX. The experimental study testified the inverse square root dependence of the localization precision on the detected photon number. It was shown that the maximum likelihood estimator attains the Cramér-Rao lower bound for 100 and more collected photons. The measurements indicate that p-MINFLUX provides at least 2 nm precision corresponding to 4-5 nm spatial resolution.

It is desirable to relate the theoretical findings of the simulations to real experiments. The dependence of the localization precision on the pixel size and the EBP misalignment can be tested by varying the respective parameters of the PSF measurement and using the fitted PSFs for the analysis of one MINFLUX measurement. Employing gold nanoparticles offers the possibility to demonstrate tracking on the nanometre scale by moving the stage according to a pre-defined trajectory.

The quality of the zero of the doughnut PSFs will benefit from further optimization of the incoming polarization state. The lengthy alignment of four doughnut PSFs will profit from the use of adaptive optics, for example, a SLM. A single SLM might be sufficient to optimize all four PSFs at once. Piezo-driven mirrors positioned in the plane conjugated to the back focal plane of the objective would permit the automated alignment of the excitation pattern creating a feedback loop to the confocal scan unit. They furthermore enable to zoom into the sample decreasing the diameter of the excitation pattern and shift it as a whole. To

reduce the number of optical components in future implementations, it might be possible to employ fibers directly creating a doughnut-shaped beam profile [109].

Future experiments involving DNA origami structures with nanometre-precise positioned fluorophores will enable the benchmarking of p-MINFLUX in terms of localization precision and spatial resolution [110, 111]. An alternative are biological structures, for example, the nuclear pore complex which serves as a reference standard in fluorescence nanoscopy [112]. In addition to the relative technical simplicity of p-MINFLUX, it provides access to lifetime information. This will open the door to super-resolved lifetime and FRET imaging, as well as lifetime-multiplexed imaging using a single wavelength [113, 114].

Based on an existing Python code, a graphical user interface was developed facilitating the data analysis process. Currently, the analysis software is extended to create a comprehensive MINFLUX analysis toolkit for the evaluation of measurement data from experiments with continuous and pulsed-interleaved excitation. The aim is to provide open-source software accessible to everyone. A modular structure allows customizing the software to the specific needs of the user. As an extension of the existing GUI, the detailed analysis of lifetime data shall be added. Users would benefit from the real-time evaluation of experimental data paving the way for many new applications [115].

7. Acknowledgement

This thesis is dedicated to my Argentinian friends who introduced me to their unique culture and most importantly helped me get through some incredibly challenging times. I cannot express how thankful I am.

I would like to thank Prof. Dr. Fernando Stefani for allowing me to work as part of his amazing research group and for his guidance - starting with the application to fund my research and including proofreading my thesis. As well as all the scientific advice you offered, I will not forget our interesting discussion about science policy. My gratitude also goes to Prof. Dr. Oliver Benson for his support of my stay abroad and critical comments on the thesis.

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Many of us are still facing a reality which seems unreal. Together we will withstand. With the help of everyone and without forgetting anyone.

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A.1. Supplementary Material for Laboratory Experiments

Table A.1.: Summary of instrument drivers for setup control by PyFLUX. Except for the auxiliary CW laser, all instruments were controlled using .dll-libraries provided by the manufacturer. Custom Python drivers were written for the indicated devices.

Device	.dll library	Custom Python driver	Source
Picoquant PH300 & PHR800	x	x	see A.3.2, [116]
DAQ board Gold II	x		[117]
Andor 887	x	x	[118]
Thorlabs ThorCam DCC1240C	x		[119, 120]
Lambda mini Evo, RGB Photonics GmbH		x	see A.3.2

Table A.2.: Polarization contrast and coupling efficiency of four excitation beams. The polarization is measured using the polarization tool described in chapter 3. The coupling efficiency of the fibers was measured comparing the laser powers before and after the respective fibre.

PSF No./ Fiber color coating	Coupling efficiency [%]	Polarization contrast [%]
1/ Orange	46.4	2.5
2/ Yellow	48.2	4.0
3/ Black	47.1	8.3
4/ Blue	53.8	9.0

Table A.3.: Excitation laser intensities. The power was measured at the back focal plane of the objective.

Excitation beam	Power [μ W] at $\lambda = 647$ nm
IR	940
Widefield	16.5
Pulsed excitation from orange fiber	9.0

Table A.4.: Background measurement. The count rates of the background were measured for the illumination with the indicated lasers.

Excitation beam(s)	Count rate [kHz]
No	0.34
IR	0.35
IR + widefield	7.22
All except red diode laser	16.98

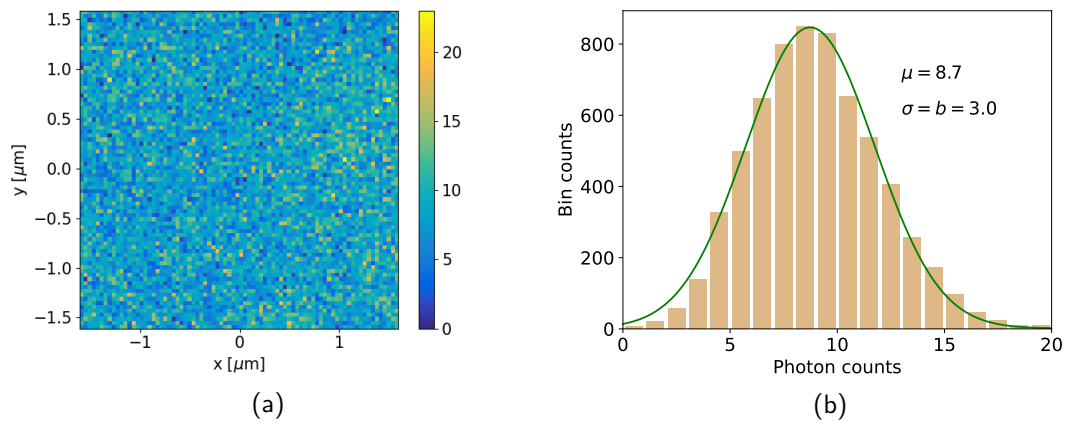


Figure A.1.: Imaging and analysis of background photon distribution. (a) Confocal image of background counts. Confocal scan with dwell time of 0.5 ms and 20 nm pixel size. (b) Histogram of measured photon distribution.

A.1.1. Protocol for Au NP + Bead Immobilization

Cleaning of cover slips:

1. Sonicate for 15min in 0.2% Hellmanex III solution (Hellma GmbH & Co. KG)
2. Wash 3× with Milli-Q H₂O (Merck KGaA)
3. Wash 3× with acetone
4. Dry

Preparation of flow chamber:

1. Fix 2 stripes of double-sided tape separated by ~ 1 cm on the longer sides of a glass side
2. Attach cleaned cover slip on top of the tape
3. Remove jutting parts of tape
4. Clean flow chamber incubating 1 M KOH for 20 min
5. Wash 3× with 100 μ L Milli-Q H₂O

Immobilization of Au NP and beads:

1. Incubate 100 μ L PDDA solution for 15 min
2. Wash 3× with 100 μ L Milli-Q H₂O
3. Incubate a solution (100 μ L) of Au NP and/ or beads for 15 min
 - $c_{\text{Au NP}} = 70\text{-}80 \%$
 - $c_{\text{Beads}} = 2 \times 10^{-5}$
 - Vortex and sonicate Au NP/ bead solution for 5 min before incubation
4. Seal flow chamber around the cover slip with Loctite 3430 epoxy adhesive (Henkel AG & Co.KGaA)

A.2. Supplementary Simulation Results

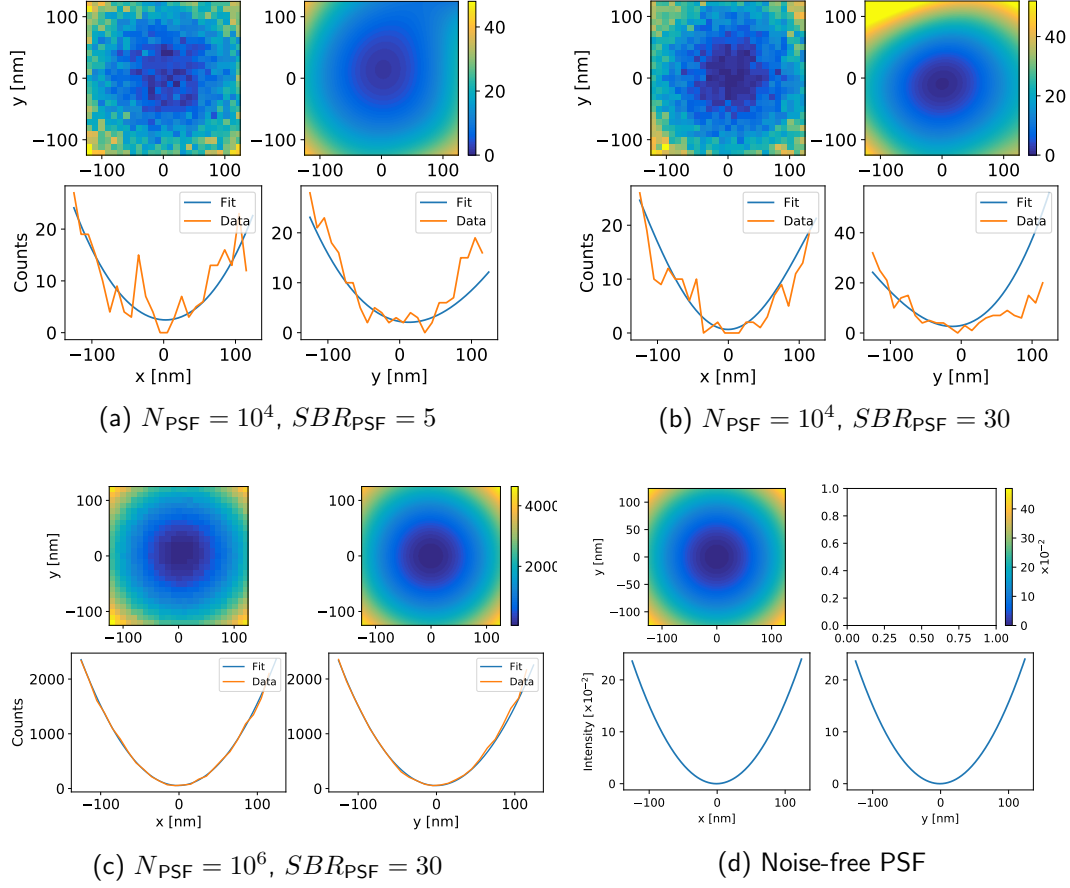


Figure A.2.: PSF simulation and fitting. The insets in (a)-(d) depict: Top-left: Simulated PSF with noise for $px_{\text{PSF}} = 10$ nm, top-right: PSF fit with $px_{\text{PSF}} = 1$ nm, bottom-left: cross-section in x, bottom-right: cross-section in y. In all cases, PSF 1 is depicted. For the noise-free PSF, the upper-left inset is blank to visualize that no fitting is required and the PSF as shown is directly evaluated.

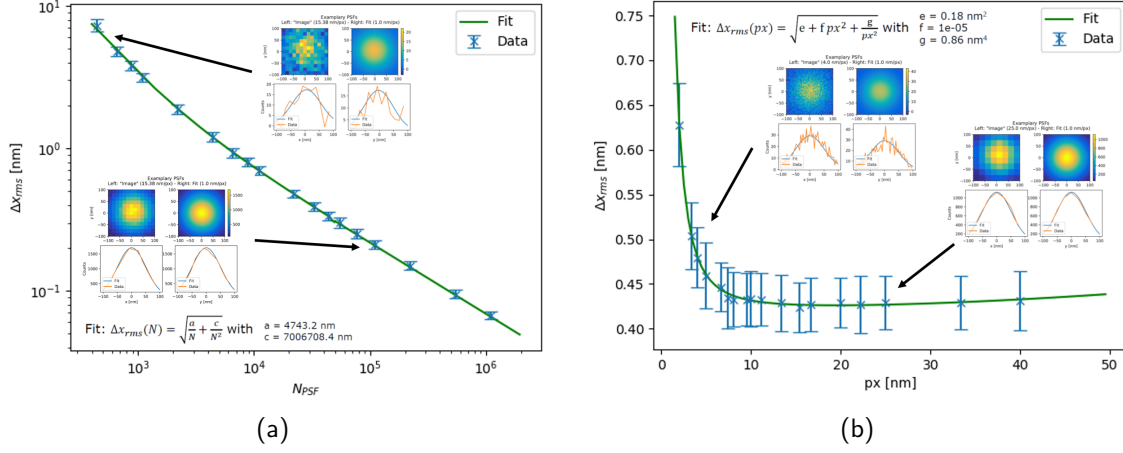


Figure A.3.: Centroid fitting of simulated emission PSFs. (a) Dependence on photon number N_{PSF} for $px_{\text{PSF}} = 15.38$ nm. (b) Dependence on pixel size px_{PSF} for $N_{\text{PSF}} = 25000$. Δx_{rms} evaluates the precision of the x-coordinate of the fitted centroid. Simulation of a Gaussian emission PSF with $\sigma_{\text{PSF}} = 50$ nm detected by a camera with $FOV_{\text{PSF}} = 200$ nm. A Gaussian background contribution was modelled according to a SBR_{PSF} of 20 and $b = 3$. Each data point is the result of 10000 simulations. The fitted dependencies of the resulting error Δx_{rms} agree with the theoretical prediction by formula 2.4.11.

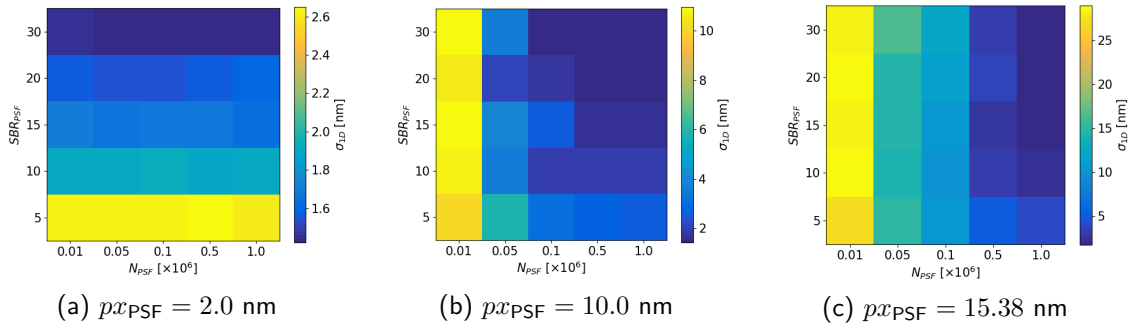


Figure A.4.: Maps of localization precision for different N_{PSF} and SBR_{PSF} combinations. For each pixel of the map, 2000 runs were carried out. $SBR_{\text{MINFLUX}} = 20$.

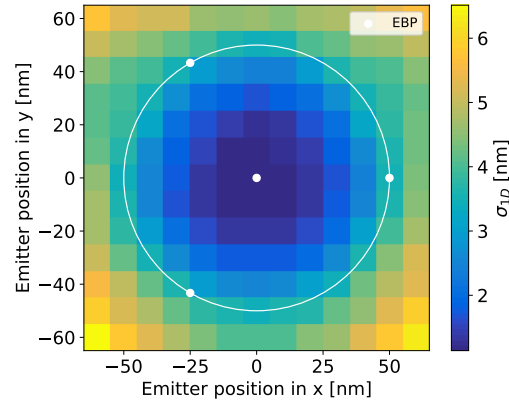


Figure A.5.: Localization precision for different emitter positions. A MINFLUX simulation was performed for $SBR_{\text{MINFLUX}} = 20$, $N_{\text{MINFLUX}} = 1000$ and noise-free PSFs. For each emitter positions, 1000 estimations were carried out.

A.3. Additional Software Information

A.3.1. Hard- and Software Specification

Table A.5.: Hard- and software configuration for Monte-Carlo simulations.

Hardware	
Graphics Card	Nvidia GeForce RTX 2070 SUPER
Graphics Card Memory	8192 MB GDDR6
CPU	AMD Ryzen 7 3700X, 8 Cores @ 3.6 GHz
RAM	32.0 GB
Software	
Operational System	Microsoft Windows 10 64-bit
Nvidia Graphics Driver Version	451.67
Python Version	Python 3.7.6
Python Distribution	Anaconda 2020.02
Python Editor	Spyder 4.1.4
CUDA Toolkit Version	10.2.89
C++ Compiler	Microsoft Visual Studio Community 2019, v16.6.4

A.3.2. Developed Code

Python Driver for TCSPC Module This custom-written driver uses the .dll library provided by PicoQuant to control the PH300 TCSPC module [116]. The driver further allows connecting to the PicoQuant PHR800 router.

Listing A.1: Custom driver for PicoQuant PH300 and PHR800.

```
1 # -*- coding: utf-8 -*-
2 """
3 Created on Tue Nov 20 14:46:50 2018
4
5 @author: Luciano A. Masullo
6 """
7 import ctypes
8 from ctypes import byref, POINTER
9 from lantz import LibraryDriver
10 from lantz import Driver, Feat, Action
11 import time
12 from datetime import datetime
13 import numpy as np
14
15 LIB_VERSION = "3.0"
16 MAXDEVNUM = 8
17 MODE_T2 = 2
18 MODE_T3 = 3
19 TTREADMAX = 131072
20 FLAG_OVERFLOW = 0x0040
21 FLAG_FIFOFULL = 0x0003
22 DEV_NUM = 0      # device number, works for only 1 PH device
23
```

```

24 class PicoHarp300(LibraryDriver):
25
26     LIBRARY_NAME = "phlib64.dll"
27
28     def __init__(self, *args, **kwargs):
29         super().__init__(*args, **kwargs)
30
31         #From phdefin.h
32         # Variables to store information read from DLLs
33         self.buffer = (ctypes.c_uint * TTREADMAX)()
34         self.libVersion = ctypes.create_string_buffer(b"", 8)
35         self.hwSerial = ctypes.create_string_buffer(b"", 8)
36         self.hwPartno = ctypes.create_string_buffer(b"", 8)
37         self.hwVersion = ctypes.create_string_buffer(b"", 8)
38         self.hwModel = ctypes.create_string_buffer(b"", 16)
39         self.errorString = ctypes.create_string_buffer(b"", 40)
40         self.res = ctypes.c_double()
41         self.countRate0 = ctypes.c_int()
42         self.countRate1 = ctypes.c_int()
43         self.flags = ctypes.c_int()
44         self.nactual = ctypes.c_int()
45         self.ctcDone = ctypes.c_int()
46         self.warnings = ctypes.c_int()
47         self.warningstext = ctypes.create_string_buffer(b"",
48                                                         16384)
49
50         # Measurement parameters,
51         #these are hardcoded since this is just a demo
52         # set T2 or T3 here,
53         #observe suitable Syncdivider and Range!
54         self.mode = MODE_T3
55         # you can change this, meaningful only in T3 mode
56         self.binning = 0
57         # you can change this, meaningful only in T3 mode
58         self.offsetValue = 0
59         # Measurement time in millisec, you can change this
60         self.tacq = 1000
61         # you can change this, observe mode! READ MANUAL!
62         self.syncDiv = 1
63         # you can change this (in mV)
64         self.CFDZeroCross0 = 10
65         # you can change this (in mV)
66         self.CFDLevel0 = 50
67         # you can change this (in mV)
68         self.CFDZeroCross1 = 10
69         # you can change this (in mV)
70         self.CFDLevel1 = 150
71         self.maxRes = 4 # max res of PicoHarp 300 in ps
72
73     def getLibraryVersion(self):
74
75         self.lib.PH_GetLibraryVersion(self.libVersion)
76
77         return self.libVersion.value.decode("utf-8")
78
79     @Action()
80     def open(self):
81

```

```

82         retcode = self.lib.PH_OpenDevice(ctypes.c_int(DEV_NUM),
83                                           self.hwSerial)
84     if retcode == 0:
85         print(datetime.now(), '[picoharp 300]' +
86               'device-number: %1d' +
87               'S/N %s initialized" % (DEV_NUM',
88               self.hwSerial.value.decode("utf-8")))
89
90     else:
91         if retcode == -1: # ERROR_DEVICE_OPEN_FAIL
92             print(datetime.now(), "[picoharp 300] %1d' +
93                   'no device" % DEV_NUM)
94         else:
95             self.lib.PH_GetErrorString(self.errorString,
96                                       ctypes.c_int(retcode))
97
98             print(datetime.now(), "[picoharp 300]' +
99                   '%1d %s" %' (DEV_NUM,
100                              self.errorString.value.decode("utf8")))
101
102     def getHardwareInfo(self):
103         self.lib.PH_GetHardwareInfo(DEV_NUM,
104                                     self.hwModel,
105                                     self.hwPartno,
106                                     self.hwVersion)
107
108         return [self.hwModel.value.decode("utf-8"),
109               self.hwPartno.value.decode("utf-8"),
110               self.hwVersion.value.decode("utf-8")]
111
112     def setup_ph300(self):
113         #setup_ph300
114         self.lib.PH_Calibrate(ctypes.c_int(DEV_NUM))
115
116         self.lib.PH_SetSyncDiv(ctypes.c_int(DEV_NUM),
117                               ctypes.c_int(self.syncDivider))
118
119         self.lib.PH_SetInputCFD(ctypes.c_int(DEV_NUM),
120                                ctypes.c_int(0),
121                                ctypes.c_int(self.CFDLevel0),
122                                ctypes.c_int(self.CFDZeroCross0))
123
124         self.lib.PH_SetInputCFD(ctypes.c_int(DEV_NUM),
125                                ctypes.c_int(1),
126                                ctypes.c_int(self.CFDLevel1),
127                                ctypes.c_int(self.CFDZeroCross1))
128
129         time.sleep(0.2)
130
131     def setup_phr800(self):
132         #seperate setup procedure for PHR800
133         #start enabling routing
134         self.lib.PH_EnableRouting(ctypes.c_int(DEV_NUM),
135                                  ctypes.c_int(1))
136
137         model = ctypes.create_string_buffer(b"", 8)
138         version = ctypes.create_string_buffer(b"", 8)
139

```

```

140         self.lib.PH_GetRouterVersion(ctypes.c_int(DEV_NUM),
141                                     byref(model),
142                                     byref(version))
143         modelstr = model.value.decode("utf-8")
144
145         if modelstr == 'PHR 800':
146             # prepare all PHR800 channels
147             for i in range(0, 4):
148 #setup phr channels and their respective levels to -200 mV
149 #no other channel level was working in order to setup the phr800
150 #according to the official software
151 #a level of 1500 mV should be adequate
152                 self.lib.PH_SetPHR800Input(ctypes.c_int(DEV_NUM),
153                                             ctypes.c_int(i),
154                                             ctypes.c_int(-200),
155                                             ctypes.c_int(0))
156                 self.lib.PH_SetPHR800CFD(ctypes.c_int(DEV_NUM),
157                                           ctypes.c_int(i),
158                                           ctypes.c_int(0),
159                                           ctypes.c_int(0))
160             else:
161                 print(datetime.now(),
162                       '[picoharp 300] No PHR800 router connected!')
163             time.sleep(0.2)
164
165     @Feat
166     def binning(self):
167         return self.binningValue
168
169     @binning.setter
170     def binning(self, value):
171         self.lib.PH_SetBinning(ctypes.c_int(DEV_NUM),
172                               ctypes.c_int(value))
173         self.binningValue = value
174
175     @Feat
176     def offset(self):
177         return self.offsetValue
178
179     @offset.setter
180     def offset(self, value):
181         self.lib.PH_SetOffset(ctypes.c_int(DEV_NUM),
182                              ctypes.c_int(value))
183         self.offsetValue = value
184
185     @Feat
186     def resolution(self):
187         self.lib.PH_GetResolution(ctypes.c_int(DEV_NUM),
188                                  byref(self.res))
189         return self.res.value
190
191     @resolution.setter
192     def resolution(self, value):
193         # calculation: resolution = maxRes * 2**binning
194         self.binning = int(np.log(value/self.maxRes)/np.log(2))
195
196     def countrate(self, channel):
197         if channel == 0:

```

```

198         self.lib.PH_GetCountRate(ctypes.c_int(DEV_NUM),
199                                   ctypes.c_int(0),
200                                   byref(self.countRate0))
201         value = self.countRate0.value
202     if channel == 1:
203         self.lib.PH_GetCountRate(ctypes.c_int(DEV_NUM),
204                                   ctypes.c_int(1),
205                                   byref(self.countRate1))
206         value = self.countRate1.value
207     return value
208
209     @Feat
210     def syncDivider(self):
211         return self.syncDiv
212
213     @syncDivider.setter
214     def syncDivider(self, value):
215         self.lib.PH_SetSyncDiv(ctypes.c_int(DEV_NUM),
216                                ctypes.c_int(value))
217         self.syncDiv = value
218
219     def startTTTR(self, outputfilename):
220         outputfile = open(outputfilename, "wb+")
221         progress = 0
222         self.lib.PH_StartMeas(ctypes.c_int(DEV_NUM),
223                               ctypes.c_int(self.tacq))
224         print(datetime.now(),
225               '[picoharp 300] TCSPC measurement started')
226 # save real time for correlating with confocal images for FLIM
227         f = open(outputfilename + '_ref_time_tcspc', "w+")
228         f.write(str(datetime.now()) + '\n')
229         f.write(str(time.time()) + '\n')
230         meas = True
231         self.measure_state = 'measuring'
232         while meas is True:
233             self.lib.PH_GetFlags(ctypes.c_int(DEV_NUM),
234                                  byref(self.flags))
235             if self.flags.value & FLAG_FIFOFULL > 0:
236                 print(datetime.now(),
237                       '[picoharp 300] FiFo Overrun!')
238                 self.stopTTTR()
239             self.lib.PH_ReadFiFo(ctypes.c_int(DEV_NUM),
240                                  byref(self.buffer),
241                                  TTREADMAX,
242                                  byref(self.nactual))
243             if self.nactual.value > 0:
244                 print(datetime.now(),
245                       '[picoharp 300] current photon count:',
246                       self.nactual.value)
247 # We could just iterate through our buffer with a for loop,
248 # however, this is slow and might cause a FIFO overrun.
249 # So instead, we shrinken
250 # the buffer to its appropriate length with array slicing,
251 # which gives us a python list.
252 # This list then needs to be converted back into
253 # a ctype array which can be written at once to the output file
254         outputfile.write((ctypes.c_uint*self.nactual.value)
255                           (*self.buffer[0:self.nactual.value]))

```

```

256         progress += self.nactual.value
257     else:
258         self.lib.PH_CTCStatus(ctypes.c_int(DEV_NUM),
259                               byref(self.ctcDone))
260         if self.ctcDone.value > 0:
261             print(datetime.now(), "[picoharp 300] Done")
262             self.numRecords = progress
263             self.stopTTTR()
264 # save real time for correlating with confocal images for FLIM
265             f.write(str(datetime.now()) + '\n')
266             f.write(str(time.time()) + '\n')
267             f.close()
268             print(datetime.now(),
269                   '[picoharp 300] {} events'
270                   + 'recorded'.format(self.numRecords))
271             meas = False
272             self.measure_state = 'done'
273
274     def stopTTTR(self):
275         self.lib.PH_StopMeas(ctypes.c_int(DEV_NUM))
276
277     def initialize(self):
278         self.lib.PH_Initialize(ctypes.c_int(DEV_NUM),
279                                ctypes.c_int(self.mode))
280
281     def finalize(self):
282         self.lib.PH_CloseDevice(ctypes.c_int(DEV_NUM))

```

Python Driver for Lambda mini Evo The Python driver establishes a connection to the CW diode laser using a serial communication protocol. A set of functions to initialize, start and control the laser power were implemented.

Listing A.2: Custom driver for RGB Lasersystems Lambda mini Evo.

```

1  # -*- coding: utf-8 -*-
2  """
3      driver for RGB Lasersystems MiniLaser Evo
4      adopted from Lantz package
5      written by Lars Richter and Luciano A. Masullo
6      date: 17.12.2019
7
8  """
9  import serial
10 MSG_SUFFIX = '\r\n'
11
12 class MiniLasEvo:
13     """Driver for any RGB Lasersystems MiniLas Evo laser.
14     """
15     def __init__(self, port):
16         self.PORT = port
17         BAUDRATE = 57600
18         BYTESIZE = 8
19         PARITY = serial.PARITY_NONE
20         STOPBITS = 1
21         self.TIMEOUT = 2 #in s, necessary for readline() call
22         #flow control flags
23         RTSCTS = False

```

```

24     DSRDTR = False
25     XONXOFF = False
26     self.serial = serial.Serial(port=self.PORT,
27                                 baudrate=BAUDRATE,
28                                 bytesize=BYTESIZE,
29                                 parity=PARITY,
30                                 stopbits=STOPBITS,
31                                 timeout=self.TIMEOUT,
32                                 xonxoff=XONXOFF,
33                                 rtscts=RTSCTS,
34                                 dsrdtr=DSRDTR)
35
36     def closeLaserPort(self):
37         self.serial.close()
38
39     def query(self, command):
40         """Send query to the laser and return the answer
41
42         :param command: command to be sent to the instrument
43         :type command: string
44         """
45         command = command + MSG_SUFFIX
46         write_return = self.serial.write(command.encode('ascii'))
47
48         code = int(write_return)
49         if code != 0:
50             if code == '1':
51                 print('[MiniLasEvo] Command invalid')
52             elif code == '2':
53                 print('[MiniLasEvo] Wrong
54                     number of parameters')
55             elif code == '3':
56                 print('[MiniLasEvo] Parameter
57                     value is out of range')
58             elif code == '4':
59                 print('[MiniLasEvo] Unlocking
60                     code is wrong')
61             elif code == '5':
62                 print('[MiniLasEvo] Device is
63                     locked for this command')
64             elif code == '6':
65                 print('[MiniLasEvo] This function is
66                     not supported')
67             elif code == '7':
68                 print('[MiniLasEvo] Timeout while
69                     reading command (60 s)')
70             elif code == '8':
71                 print('[MiniLasEvo] This value is
72                     currently not available')
73
74         message = self.serial.readline().decode()
75         message = message.replace('0 ', '')
76         message = message.replace(MSG_SUFFIX, '')
77         return message
78
79     def idn(self):
80         """Identification of the device
81         """

```

```

82     manufacturer = self.query('DM?')
83     device = self.query('DT?')
84     serialnum = self.query('DS?')
85     ans = manufacturer + ', ' + device
86         + ', serial number: ' + serialnum
87     return ans
88
89 def status(self):
90     """Current device status"""
91
92     ans = self.query('S?')
93
94     ans1 = ''
95     ans2 = ''
96
97     if ans[-2] == '1':
98         ans1 = 'Temperature of laser head is ok'
99     if ans[-1] == '1':
100         ans2 = 'Laser system is active,
101             radiation can be emitted'
102     elif ans[-1] == '2':
103         ans2 = '(reserved)'
104     elif ans[-1] == '4':
105         ans2 = 'The interlock is open'
106     elif ans[-1] == '8':
107         ans = self.query('E?')
108         if ans[-1] == '1':
109             ans2 = 'Temperature of laser head is too high'
110         elif ans[-1] == '2':
111             ans2 = 'Temperature of laser head is too low'
112         elif ans[-1] == '4':
113             ans2 = 'Temperature-sensor connection is broken'
114         elif ans[-1] == '8':
115             ans2 = 'Temperature sensor cable is shortened'
116
117         if ans[-2] == '4':
118             ans1 = 'Current for laser head is too high'
119         elif ans[-2] == '8':
120             ans1 = 'Internal error
121                 (laser system cannot be activated)'
122
123     return ans1, ans2
124
125 def operating_hours(self):
126     """Total operating hours [hhhh:mm]"""
127
128     return self.query('R?')
129
130 def software_version(self):
131     """Software version"""
132
133     return self.query('DO?')
134
135 def emission_wavelength(self):
136     """Emission wavelength in nm"""
137
138     return self.query('DW?')
139

```

```

140     def available_features(self):
141         """Available features (reserved for future use)"""
142         """
143         return self.query('DF?')
144
145     def control_mode(self):
146         """Active current (power) control"""
147         """
148         ans = self.query('DC?')
149         if ans == 'ACC':
150             ans = 'Active current control'
151         else:
152             ans = 'Active power control'
153         return ans
154
155     # TEMPERATURE
156     def temperature(self):
157         """Current temperature in deg"""
158         """
159         return self.query('T?')
160
161     def temperature_min(self):
162         """Lowest operating temperature in deg"""
163         """
164         return self.query('LTN?')
165
166     def temperature_max(self):
167         """Highest operating temperature in deg"""
168         """
169         return self.query('LTP?')
170
171     @property
172     def enabled(self):
173         """Method for turning on the laser"""
174         """
175         return bool(int(self.query('O?')))
176
177     @enabled.setter
178     def enabled(self, value):
179         """Method for turning on the laser"""
180         value: True, False
181         """
182         value = str(int(value))
183         self.query('O=' + value)
184
185     def initialize(self):
186         super().initialize()
187         self.feats.power.limits = (0, self.maximum_power.magnitude)
188
189     def maximum_power(self):
190         """Gets the maximum emission power of the laser"""
191         """
192         return float(self.query('LP?'))
193
194     @property
195     def power(self):
196         """Gets and sets the emission power"""
197         """

```

```

198         return float(self.query('P?'))
199
200     @power.setter
201     def power(self, value):
202         self.query('P={:.1f}'.format(value))
203
204 if __name__ == '__main__':
205
206     port = 'COM7'
207     minilaser = MiniLasEvo(port)

```

CUDA C++ Code describing 2D-Polynomial C++ script extending the Gpufit package to fit 2D-polynomials.

Listing A.3: C++ code defining 2D-polynomial for fit by Gpufit.

```

1 #ifndef GPUFIT_POLYNOM2D_CUH_INCLUDED
2 #define GPUFIT_POLYNOM2D_CUH_INCLUDED
3
4 /* Description of the calculate_polynomial2d function
5 * =====
6 *
7 * This function calculates the values of a two-dimensional polynomial
8 * Also, it states its partial derivatives
9 *
10 * No independent variables are passed to this model function. Hence, the
11 * (X, Y) coordinate of the first data value is assumed to be (0.0, 0.0).
12 For
13 * a fit size of M x N data points, the (X, Y) coordinates of the data are
14 * simply the corresponding array index values of the data array, starting
15 from
16 * zero.
17 *
18 * Parameters:
19 *
20 * parameters: An input vector of model parameters.
21 * p[0]: x0
22 * p[1]: y0
23 * p[2]: c00
24 * p[3]: c01
25 * p[4]: c02
26 * p[5]: c03
27 * p[6]: c04
28 * p[7]: c10
29 * p[8]: c11
30 * p[9]: c12
31 * p[10]: c13
32 * p[11]: c14
33 * p[12]: c20
34 * p[13]: c21
35 * p[14]: c22
36 * p[15]: c23
37 * p[16]: c30
38 * p[17]: c31
39 * p[18]: c32
40 * p[19]: c33
41 * p[20]: c40
42 * p[21]: c41

```

```

41 *
42 * n_fits: The number of fits. (not used)
43 *
44 * n_points: The number of data points per fit.
45 *
46 * value: An output vector of model function values.
47 *
48 * derivative: An output vector of model function partial derivatives.
49 *
50 * point_index: The data point index.
51 *
52 * fit_index: The fit index. (not used)
53 *
54 * chunk_index: The chunk index. (not used)
55 *
56 * user_info: An input vector containing user information. (not used)
57 *
58 * user_info_size: The size of user_info in bytes. (not used)
59 *
60 * Calling the calculate_gauss2drotated function
61 * =====
62 *
63 * This __device__ function can be only called from a __global__ function
   or an other
64 * __device__ function.
65 *
66 */
67
68 __device__ void calculate_polynomial2d(
69     REAL const * parameters,
70     int const n_fits,
71     int const n_points,
72     REAL * value,
73     REAL * derivative,
74     int const point_index,
75     int const fit_index,
76     int const chunk_index,
77     char * user_info,
78     std::size_t const user_info_size)
79 {
80     // indices
81
82     int const n_points_x = sqrt((REAL)n_points);
83
84     int const point_index_y = point_index / n_points_x;
85     int const point_index_x = point_index - point_index_y * n_points_x;
86
87     // parameters
88
89     REAL const * p = parameters;
90
91     // value
92
93     REAL const x = point_index_x - p[0];
94     REAL const xp2 = x * x;
95     REAL const xp3 = xp2 * x;
96     REAL const xp4 = xp3 * x;
97     REAL const y = point_index_y - p[1];

```

```

98     REAL const yp2 = y * y;
99     REAL const yp3 = yp2 * y;
100    REAL const yp4 = yp3 * y;
101
102    REAL const sum0 = p[2] + p[3] * y + p[4] * yp2 + p[5] * yp3 + p[6] *
yp4;
103    REAL const sum1 = x * (p[7] + p[8] * y + p[9] * yp2 + p[10] * yp3 + p
[11] * yp4);
104    REAL const sum2 = xp2 * (p[12] + p[13] * y + p[14] * yp2 + p[15] *
yp3);
105    REAL const sum3 = xp3 * (p[16] + p[17] * y + p[18] * yp2 + p[19] *
yp3);
106    REAL const sum4 = xp4 * (p[20] + p[21] * y);
107
108    value[point_index] = sum0 + sum1 + sum2 + sum3 + sum4;
109
110    // derivative
111
112    REAL * current_derivative = derivative + point_index;
113
114    REAL const dxsum0 = -1 * (p[7] + p[8] * y + p[9] * yp2 + p[10] * yp3
+ p[11] * yp4);
115    REAL const dxsum1 = -2 * x * (p[12] + p[13] * y + p[14] * yp2 + p[15]
* yp3);
116    REAL const dxsum2 = -3 * xp2 * (p[16] + p[17] * y + p[18] * yp2 + p
[19] * yp3);
117    REAL const dxsum3 = -4 * xp3 * (p[20] + p[21] * y);
118    current_derivative[0 * n_points] = dxsum0 + dxsum1 + dxsum2 + dxsum3;
119
120    REAL const dysum0 = -1 * (p[3] + p[8] * x + p[13] * xp2 + p[17] * xp3
+ p[21] * xp4);
121    REAL const dysum1 = -2 * y * (p[4] + p[9] * x + p[14] * xp2 + p[18] *
xp3);
122    REAL const dysum2 = -3 * yp2 * (p[5] + p[10] * x + p[15] * xp2 + p
[19] * xp3);
123    REAL const dysum3 = -4 * yp3 * (p[6] + p[11] * x);
124    current_derivative[1 * n_points] = dysum0 + dysum1 + dysum2 + dysum3;
125
126    current_derivative[2 * n_points] = 1;
127    current_derivative[3 * n_points] = y;
128    current_derivative[4 * n_points] = yp2;
129    current_derivative[5 * n_points] = yp3;
130    current_derivative[6 * n_points] = yp4;
131    current_derivative[7 * n_points] = x;
132    current_derivative[8 * n_points] = x * y;
133    current_derivative[9 * n_points] = x * yp2;
134    current_derivative[10 * n_points] = x * yp3;
135    current_derivative[11 * n_points] = x * yp4;
136    current_derivative[12 * n_points] = xp2;
137    current_derivative[13 * n_points] = xp2 * y;
138    current_derivative[14 * n_points] = xp2 * yp2;
139    current_derivative[15 * n_points] = xp2 * yp3;
140    current_derivative[16 * n_points] = xp3;
141    current_derivative[17 * n_points] = xp3 * y;
142    current_derivative[18 * n_points] = xp3 * yp2;
143    current_derivative[19 * n_points] = xp3 * yp3;
144    current_derivative[20 * n_points] = xp4;
145    current_derivative[21 * n_points] = xp4 * y;

```

```
146 }  
147  
148 #endif
```

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B. Selbstständigkeitserklärung

Ich erkläre hiermit, dass ich die vorliegende Arbeit selbstständig verfasst und noch nicht für andere Prüfungen eingereicht habe. Sämtliche Quellen einschließlich Internetquellen, die unverändert oder abgewandelt wiedergegeben werden, insbesondere Quellen für Texte, Grafiken, Tabellen und Bilder, sind als solche kenntlich gemacht. Mir ist bekannt, dass bei Verstößen gegen diese Grundsätze ein Verfahren wegen Täuschungsversuchs bzw. Täuschung eingeleitet wird.

Fredersdorf, d. Richter